

Language Equivalence Between Deterministic One-Counter Automata and Nondeterministic One-Counter Nets Is Decidable

The Real-Time Case and the Epsilon-Move Extension

Alp Eren Bütün
Independent Researcher, Türkiye
ORCID: 0009-0001-2479-6037
github.com/alperenbutun

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Abstract

We settle language equivalence between a deterministic zero-testing one-counter automaton and a nondeterministic one-counter net in the standard real-time model, thereby giving a decidability theorem for the real-time formulation recorded as Problem 24.04(A) in the Automata Exchange collection. We then prove a stronger extension in which both machines may have silent transitions: the deterministic side uses the standard deterministic-epsilon convention and the nondeterministic net may have arbitrary epsilon moves.

The difficult inclusion is from the deterministic zero-testing machine into the nondeterministic net. In the real-time case we develop exact forward-envelope/backward-requirement semantics, decompose a shortest counterexample at deterministic zero cuts, prove an effective node-dependent compactness theorem for positive first-return corridors, and assemble the result through an effectively finite zero-state tree and a terminal drain reduction. For the epsilon extension we derive a finite weak-envelope normal form: weak visible steps are represented by finitely many threshold-affine macros together with explicit pumping thresholds and an omega value for unbounded credit. A guarded-macro transfer theorem proves that the real-time compactness architecture is robust under this compilation; threshold contacts, pump obstructions, omega-regime refinement, and concrete weak skeletons preserve the rank and literal-trace interfaces of the compactness proof. Reset-form deterministic silent behavior is then incorporated into the zero-return decomposition.

In both models the reverse inclusion reduces to exact reachability in a vector addition system with one zero test. Consequently language equality is decidable in the real-time model and remains decidable under the stated epsilon-transition conventions. No useful complexity upper bound is claimed.

Keywords. one-counter automata; one-counter nets; language equivalence; epsilon transitions; zero tests; residual semantics; compactness; well-quasi-orders; decidability.

1 Introduction

A one-counter automaton (OCA) is a pushdown automaton with a singleton stack alphabet. Unlike a one-counter net (OCN), it may distinguish counter value 0 from positive values. We study the asymmetric language-equivalence problem in which the zero-testing machine is deterministic and the net is nondeterministic.

Problem 24.04(A) in the Automata Exchange collection asks whether language equivalence between a deterministic one-counter automaton and a nondeterministic one-counter net is decidable [1]. The letter-labelled formalization used in the surrounding one-counter-net literature

is real time; we first solve that formulation and then prove that the same decidability survives a standard deterministic-epsilon extension on the OCA side together with arbitrary epsilon moves on the OCN side.

Theorem 1.1 (Real-time main theorem). *Let A be a deterministic real-time unit-step one-counter automaton with zero tests and let N be a nondeterministic real-time unit-step one-counter net over the same finite alphabet. Both start with counter value 0 and accept finite words by final control state. It is decidable whether $L(A) = L(N)$.*

Theorem 1.2 (Real-time global small witness). *There is a total computable function B such that for every pair (A, N) as in Theorem 1.1,*

$$L(A) \not\subseteq L(N) \implies \exists w \in L(A) \setminus L(N) \text{ with } |w| \leq B(A, N).$$

Theorem 1.3 (Epsilon-move extension). *Given a deterministic unit-update zero-testing one-counter automaton A with standard deterministic epsilon moves and a nondeterministic unit-update one-counter net N with arbitrary epsilon moves and no zero tests, both initialized at counter 0 and using finite-word final-state acceptance, it is decidable whether $L(A) = L(N)$.*

Theorem 1.4 (Epsilon hard inclusion). *For the machines of Theorem 1.3, it is decidable whether $L(A) \subseteq L(N)$. Moreover a finite bound $B_{\text{global}}^\epsilon(A, N)$ is effectively computable such that failure of inclusion has a witness of visible length at most this bound.*

Theorem 1.5 (Epsilon reverse inclusion). *For the machines of Theorem 1.3, it is decidable whether $L(N) \subseteq L(A)$.*

Proof architecture. The real-time proof has four layers: exact residual semantics, positive first-return compactness, an effectively finite zero-state tree, and a terminal drain reduction. The epsilon extension does not eliminate silent moves syntactically. Instead it compiles each weak visible OCN step into a finite threshold-affine macro system, proves an internal guarded-macro robustness theorem for the compactness argument, and combines this with the standard stable/reset normalization of deterministic epsilon-OCA. The reverse inclusion in both settings is handled independently by one-zero-test VASS reachability [10, 11].

Relation to known results and novelty scope. Valiant and Paterson proved decidability of equivalence for deterministic one-counter automata [2], and Böhm, Göller, and Jančar later obtained the NL-completeness bound in the real-time setting [5]. This should be contrasted with deterministic-OCA language inclusion, which is already undecidable [3]; hence the present equality problem does not follow from a two-way inclusion test on deterministic OCAs.

On the net side, trace inclusion from a nondeterministic OCN into a deterministic OCN is decidable (indeed NL-complete in the standard trace setting), whereas general OCN trace inclusion is undecidable [7]; strong and weak simulation over OCNs remain decidable and PSPACE-complete [6]. General OCN determinizability is undecidable under most natural notions [8]. History-determinism gives a different positive frontier: history-deterministic OCNs can be converted to language-equivalent deterministic OCAs and their language inclusion/equivalence problems are decidable in PSPACE [9]. The same work notes that its HD-OCN results extend, with weak-simulation machinery, to epsilon-transitions; thus silent moves themselves are not the novelty claimed here.

The comparison class in this paper is different from all of the preceding settings: the right-hand machine is an arbitrary nondeterministic OCN, not assumed deterministic or history-deterministic, while the left-hand machine may test zero and is deterministic. To the best of our knowledge, we are not aware of a prior published decidability theorem for finite-word final-state language equality for this exact asymmetric pair, either in the real-time formulation or under

the epsilon conventions of Theorem 1.3. We cite the Automata Exchange entry only as a record of Problem 24.04(A), not as evidence of priority or novelty [1]. Problem 24.04(B), the separate net/net variant recorded on the same exchange, is not used in any argument here.

Organization. Sections 3–10 prove the real-time theorem. The subsequent epsilon-extension sections develop weak envelopes, guarded macros, reset-ending returns, the stable-zero tree, terminal tails, and the asynchronous one-zero-test product. The appendices record both effectivity ledgers and the module-by-module transfer certificate.

2 Machines and acceptance

2.1 Deterministic one-counter automata

A *real-time unit-step one-counter automaton* is a tuple

$$A = (Q_A, \Sigma, \delta_A, q_A^0, F_A),$$

where the counter ranges over $\mathbb{N}$. Transitions are separated into zero and positive rules. A zero rule is enabled exactly when the current counter is 0; a positive rule is enabled exactly when the current counter is positive. Each rule reads one input letter and changes the counter by -1 , 0 , or $+1$, subject to nonnegativity. We assume that A is deterministic: for each control state, input letter, and zero/positive regime, at most one transition is enabled.

A configuration is $(p, c) \in Q_A \times \mathbb{N}$. The initial configuration is $(q_A^0, 0)$, and a word is accepted when the unique run is defined and ends in a control state in F_A , with arbitrary final counter value.

2.2 Nondeterministic one-counter nets

A *one-counter net* is a tuple

$$N = (Q_N, \Sigma, \delta_N, q_N^0, F_N)$$

with transitions (q, a, d, r) , $d \in \{-1, 0, +1\}$. A transition is enabled at counter value c iff $c + d \geq 0$. There is no zero test. Acceptance is again by final control state with arbitrary final counter value.

For a word w , we write $L(A)$ and $L(N)$ for the corresponding final-state languages from initial counter 0.

2.3 Model scope and normalization

The theorem is stated for the standard *real-time* labelled one-counter model: every transition consumes exactly one input symbol. This is the convention used throughout the proof because literal factors of a shortest counterexample are repeatedly deleted and replaced. The Automata Exchange record for Problem 24.04(A) describes the left machine as “a pushdown automaton with one stack alphabet” and the right machine as a one-counter automaton with no zero tests, but does not itself print a transition syntax. The formal OCA/OCN definition used by Prakash and Thejaswini [9] matches the present convention: OCA transitions are given by

$$Q \times \{\text{zero}, \text{nonzero}\} \times \Sigma \times \{-1, 0, 1\} \times Q,$$

and OCN transitions by $Q \times \Sigma \times \{-1, 0, 1\} \times Q$, with determinism a partial function on the letter-labelled OCA rules. Their surrounding generic LTS preliminaries allow ε -labels, and one displayed OCA sentence inherits the notation Σ_ε ; however the OCA transition relation itself is letter-labelled over Σ , and their paper explicitly distinguishes realtime LTSs. We therefore treat Theorem 1.1 as the standard real-time formulation of Problem 24.04(A). The stronger

convention with silent transitions is treated separately later in the paper: the deterministic side is first put into stable/reset normal form, while arbitrary silent behavior of the net is compiled into a weak threshold-affine macro calculus.

The unit-step assumption is inessential for finite integer weights.

Proposition 2.1 (Integer-weight normalization). *Let M be a deterministic real-time one-counter automaton, or a nondeterministic real-time one-counter net, whose counter updates belong to a finite subset of $\mathbb{Z}$. One can effectively construct a language-equivalent unit-step machine of the same deterministic/nondeterministic and zero-testing/net type.*

Proof. Choose an integer K larger than every absolute transition weight and represent a counter value uniquely as $n = Kc + r$ with $0 \leq r < K$. Store r in finite control and use c as the physical counter. For a transition of weight z , write $r + z = Kk + r'$ with $k \in \{-1, 0, 1\}$; update the physical counter by k and the finite remainder to r' . Underflow of the physical counter when $k = -1$ is exactly original underflow. For an OCA, original zero is characterized by the conjunction $c = 0$ and $r = 0$, which is expressible because r is finite control and $c = 0$ is the unique zero test. If $r > 0$, the original counter is positive irrespective of c , so the corresponding original positive rule is copied into both physical zero and physical positive regimes; if $r = 0$, it is copied only to the physical positive regime. Determinism is preserved because the remainder refines the source state. For an OCN no zero test is introduced. Acceptance controls are copied for every remainder. $\square$

Remark 2.2 (Scope of Theorem 1.1). Consequently the real-time theorem covers arbitrary finite integer transition weights. Silent transitions are not folded into this normalization; they are handled by the separate epsilon-extension machinery developed later in the paper.

3 Exact residual semantics of the net

The proof uses exact finite-dimensional summaries of the nondeterministic net, rather than a selected run.

Definition 3.1 (Forward envelope). For a prefix u and $q \in Q_N$, define

$$H_u(q) = \max\{c \in \mathbb{N} : (q_N^0, 0) \xrightarrow{u} (q, c)\},$$

with $H_u(q) = \perp$ if q is unreachable. We order $\perp < h$ for every $h \in \mathbb{N}$.

Definition 3.2 (Backward requirement). For a suffix v and $q \in Q_N$, define

$$R_v(q) = \min\{c \in \mathbb{N} : \exists f \in F_N, (q, c) \xrightarrow{v} (f, c')\},$$

with $R_v(q) = \infty$ if no such counter value exists.

Proposition 3.3 (Exact cut semantics). *For every factorization $w = uv$,*

$$w \in L(N) \iff \exists q \in Q_N : H_u(q) \geq R_v(q).$$

Hence

$$w \notin L(N) \iff H_u(q) < R_v(q) \text{ for all } q \in Q_N.$$

Proof. If w is accepted, cut an accepting run after u . If the cut configuration is (q, c) then $c \leq H_u(q)$ and $c \geq R_v(q)$. Conversely, if $H_u(q) \geq R_v(q)$, choose a prefix run realizing $H_u(q)$ and use monotonicity of the net to execute an accepting v -run that needs only $R_v(q)$ units of credit. $\square$

For a fixed word x , let T_x denote the exact forward-envelope transformer and S_x the exact backward-requirement transformer.

Theorem 3.4 (Exact residuation). *For every forward envelope H , backward requirement R , and word x ,*

$$T_x(H) < R \iff H < S_x(R)$$

componentwise.

Proof. Fix a source state q . The inequality $H(q) < S_x(R)(q)$ says that every x -path starting at q either requires more initial counter than $H(q)$ or exits below the corresponding requirement in R . This is exactly the statement that no contribution from q can make $T_x(H)$ meet R . Taking all source and target states gives the equivalence. $\square$

4 Shortest counterexamples and zero cuts

Assume $L(A) \not\subseteq L(N)$ and choose a shortest word

$$w \in L(A) \setminus L(N).$$

Let (p_t, d_t) be the unique A -configuration after the prefix of length t , and let H_t and R_t be the exact net residuals at that cut. Proposition 3.3 gives

$$H_t < R_t$$

at every cut.

Lemma 4.1 (Zero-cut deletion). *Suppose $i < j$, $d_i = d_j = 0$, $p_i = p_j$, and $H_i \leq H_j$ componentwise. Then deleting the factor between cuts i and j produces a shorter word in $L(A) \setminus L(N)$.*

Proof. The deterministic run rejoins the identical physical configuration $(p_i, 0) = (p_j, 0)$, so the unchanged suffix remains accepting. At cut j , rejection gives $H_j < R_j$. Since $H_i \leq H_j$, the unchanged suffix also rejects from H_i . $\square$

Thus the zero-cut states (p_t, H_t) of a shortest counterexample form a bad sequence under equality of p and componentwise order on H .

Between consecutive zero cuts, a nontrivial factor has the form

$$0 \xrightarrow{+1} \text{strictly positive interior} \xrightarrow{-1} 0.$$

The interior contains no zero test. We call such a factor a *positive first return*.

5 Positive first-return compactness

This section contains the main technical theorem. We state it in a suffix-parametric form because the proof is local to one exact zero entry and uses the future suffix only through its exact backward requirement. This stronger formulation will also be needed for reset-ending returns in the epsilon extension.

Definition 5.1 (Residual-minimal first return). Fix a concrete exact zero entry (C_0, H_0) , a deterministic zero exit configuration C_1 , and a backward requirement vector

$$R \in (\mathbb{N} \cup \{\infty\})^{Q_N}.$$

A positive first-return word P from C_0 to C_1 is *R -minimal* if its exact exit envelope H_P satisfies $H_P < R$ and there is no strictly shorter word Q that takes A from C_0 to the same physical exit configuration C_1 and whose exact exit envelope satisfies $H_Q < R$. The competitor Q need not itself be a first return.

Theorem 5.2 (Suffix-parametric positive first-return compactness). *Fix a concrete exact zero-cut entry consisting of an A zero configuration C_0 and an exact net envelope H_0 . There is a total computable integer*

$$B_{\text{FR}}(A, N, C_0, H_0)$$

such that, for every deterministic zero exit configuration C_1 and every backward requirement vector R , every R -minimal positive first-return word from (C_0, H_0) to C_1 has literal length at most B_{FR} . In particular, every positive first-return corridor occurring from this entry inside a globally shortest word of $L(A) \setminus L(N)$ has length at most the same bound.

Indeed, if a corridor P occurs in a globally shortest counterexample and R is the exact backward requirement of the unchanged suffix at its exit, then any shorter competitor Q from the same entry to the same physical zero exit with $H_Q < R$ would splice with that suffix and produce a shorter global counterexample. Thus P is R -minimal. Conversely, the proof below uses global shortestness only through this residual-minimality implication. Throughout this section, “shortest witness” may therefore be read as “ R -minimal witness” for one fixed but arbitrary exit requirement R .

5.1 Replacement principles and anchored shells

Two different operations occur in the proof and must not be conflated.

Definition 5.3 (Trace-preserving deletion). A trace-preserving deletion removes one or more literal contiguous blocks from the *observed* R -minimal word. Every removed block is identified at its actual location in that word. The deterministic run on the remaining letters must be checked directly.

Definition 5.4 (Endpoint replacement). An endpoint replacement substitutes an observed factor P by a possibly different word Q . It is valid when Q starts at the same cut, reaches the identical deterministic A exit configuration, and its exact N exit envelope H_Q satisfies $H_Q \leq H_P$.

Lemma 5.5 (Extensional splice principle). *Let P and Q start at the same exact entry and reach the same deterministic exit configuration, with $H_Q \leq H_P$. For every backward requirement vector R ,*

$$H_P < R \implies H_Q < R.$$

Hence an R -minimal witness admits no strictly shorter endpoint-dominating replacement. Applied with R equal to the exact requirement of an actual suffix, this is precisely the usual splice argument in a globally shortest counterexample.

Proof. Componentwise $H_Q \leq H_P < R$ immediately gives $H_Q < R$. The deterministic endpoint is identical, so the same future deterministic computation is available when R comes from a concrete suffix. $\square$

Choose a peak cut z of a positive first return. For each level h below the peak, let u_h be the last height- h cut before z and v_h the first height- h cut after z . Then

$$u_h < u_k < z < v_k < v_h \quad (h < k).$$

The inner factor from u_k to v_k stays at height at least k and can therefore be translated downward by $k - h$ without activating a zero rule.

For every finite envelope coordinate put

$$F_t(q) = H_t(q) - d_t.$$

For every fixed OCN word transformer T_x and $c \geq 0$,

$$T_x(H + c\mathbf{1}) \geq T_x(H) + c\mathbf{1}. \tag{1}$$

Lemma 5.6 (Anchored shell deletion). *Let $1 \leq h < k$. Assume the corresponding ascent and descent boundary controls agree and*

$$F_{u_h} \leq F_{u_k}, \quad F_{v_k} \leq F_{v_h}.$$

Then translating the inner shell to the outer level gives a strictly shorter word reaching the same deterministic exit and still rejected against the fixed requirement R , contradicting R -minimality.

Proof. The translated factor follows the same positive-regime deterministic control path and has enough vertical reserve, hence rejoins the exact deterministic outer exit. The first anchored inequality gives no larger effective input envelope after lowering the inner shell; applying the same literal inner word and (1), then using the second anchored inequality, yields an outer exit envelope no larger than the original one. Lemma 5.5 finishes the shortening. $\square$

Thus a deep deletion-free tower forces noncomparability of anchored endpoint data. All subsequent machinery exists to turn that noncomparability into either lower-rank escape or a bounded context.

5.2 Finite owner/provenance refinement and macro systems

The later guard and wall constructions use a finite owner-resolved macro system. We make that object explicit here.

Definition 5.7 (Bounded macro type). Fix a positive-mode proof state. A *bounded macro type* records a fixed finite literal transition skeleton between two refined cuts: its source and target deterministic controls, its literal input word, its deterministic counter effect and internal entry requirement, and the finite family of N -paths over that word between states of the current finite support. Low counter values at which one of those paths changes feasibility are unfolded into finite control. At every proof state only finitely many bounded macro types are used.

For a macro e from support S to support S' , exact max-envelope propagation has the form

$$H'(r) = \max_{q \in S} \{H(q) + w_e(q, r) : H(q) \geq \lambda_e(q, r)\}, \quad r \in S',$$

where $w_e(q, r)$ and the finite entry requirement $\lambda_e(q, r)$ are effectively computable from the fixed literal macro. Fix once and for all an order on Q_N and break ties by the least maximizing predecessor.

Definition 5.8 (Owner frame and provenance color). At an owner-resolved macro cut, the *owner map*

$$\sigma_e : S' \rightarrow S$$

sends each target coordinate to its canonical maximizing predecessor. The *owner frame* records the current support, the owner map on live coordinates, and—after rank stabilization—the induced permutation of live provenance classes. Starting from a concrete entry support S_0 , give every $q \in S_0$ a distinct provenance atom $[q]$. The composed map

$$\text{Prov}_t : S_t \rightarrow S_0$$

assigns to every live coordinate at cut t the entry atom obtained by composing the intervening owner maps. A *provenance color* records the current support, the partition of current coordinates by equality of Prov_t , and the finite permutation/frame data.

All these colors are finite: Q_N is finite, the macro alphabet is finite, there are finitely many supports, maps between finite supports, set partitions, threshold regimes, and permutations. On one fixed owner-resolved edge the chosen maximizing predecessor is exact, so the live owner values satisfy an affine update

$$x' = P_e x + c_e, \quad (2)$$

with P_e the 0–1 matrix induced by the owner map. The validity of that owner choice is exactly a finite conjunction of affine difference inequalities comparing the selected predecessor with all competitors. Thus the refinement is an exact encoding of the envelope, not a selected-run approximation.

Definition 5.9 (Provenance rank). The provenance rank at a refined cut is

$$\nu(t) := |\text{im Prov}_t|.$$

A *provenance-rank drop* means that one entry provenance atom represented before a macro step is no longer represented afterwards. A raw support change that merely creates or removes coordinates carrying already represented atoms is a finite-mode change, not a rank drop.

Lemma 5.10 (Finite-depth provenance descent). *Along owner-resolved macro cuts, ν is monotone nonincreasing. If the induced owner action on the live provenance classes is non-surjective, then ν decreases strictly. Hence there are at most $|Q_N| - 1$ strict provenance-rank drops before the rank stabilizes; after stabilization every owner update induces a permutation of the live provenance classes.*

Proof. For consecutive macro cuts $t \rightarrow t'$, the provenance map factors as

$$S_{t'} \xrightarrow{\sigma_e} S_t \xrightarrow{\text{Prov}_t} S_0,$$

so $\text{im Prov}_{t'} \subseteq \text{im Prov}_t$. Strict loss of a represented class gives strict inequality of cardinalities. If the cardinality is stable, the induced map on the finite set of live provenance classes is surjective and therefore bijective. Recording that bijection is the finite permutation-frame refinement. $\square$

Proposition 5.11 (Fixed-route terminal module). *Fix one owner-resolved bounded-expression macro schema*

$$\alpha_0 \beta_1^{n_1} \alpha_1 \cdots \beta_m^{n_m} \alpha_m$$

from a concrete exact entry to one exact deterministic target configuration. Then feasibility of the schema is an effective Presburger condition on $(n_1, \dots, n_m)$, the endpoint owner vector is affine in the exponents, and the componentwise-minimal reachable endpoint set together with shortest representatives is effectively computable. Consequently this fixed-route family is shortest-witness compact.

Proof. After the finite owner/mode refinement, each powered block β_i begins and ends in the same refined mode and acts on its live owner vector by an affine map

$$x \mapsto P_i x + c_i,$$

where P_i is a 0–1 functional matrix induced by the owner map. The finite monoid generated by P_i is effectively eventually periodic: compute μ_i, λ_i with $P_i^{\mu_i + \lambda_i} = P_i^{\mu_i}$. Refining the exponent by the finitely many cases

$$n_i < \mu_i \quad \text{or} \quad n_i = \mu_i + s_i + \lambda_i k_i, \quad 0 \leq s_i < \lambda_i,$$

makes the endpoint vector affine in the quotient variable k_i . Indeed

$$P_i^{n_i} x + \sum_{j < n_i} P_i^j c_i$$

has a fixed residue-dependent linear part plus k_i times one fixed period-sum vector.

Every threshold guard, owner comparison, and deterministic-positivity condition at an intermediate repetition is an affine inequality in the current owner vector. After the same residue refinement its truth at repetition j is therefore a Presburger formula in the exponent quotients and the bounded repetition index. Requiring it at every repetition is still Presburger, since Presburger arithmetic is effectively closed under bounded and unbounded first-order quantification. The exact deterministic target is handled in the same way. Taking the finite disjunction over all residue branches yields an effective Presburger description of the feasible exponent family, and on each branch the endpoint owner vector is affine in the quotient variables (equivalently Presburger-piecewise affine in the original exponents).

Presburger arithmetic therefore decides exact endpoint slices and computes the finite semilinear feasible exponent family. Recursively minimize endpoint coordinates using exact-slice queries, exactly as in Proposition 5.17 below, and enumerate shortest realizing words. Endpoint domination then invokes Lemma 5.5. $\square$

5.3 Exact projective modes and finite-projective compression

Let $E = \text{supp}(H)$ be the finite support of an exact envelope. If $E \neq \emptyset$, define its common scalar and normalized profile by

$$z(H) = \min_{q \in E} H(q), \quad \theta(H) = H - z(H)\mathbf{1}_E.$$

Thus $\min \theta(H) = 0$. A *projective mode* records the finite data that determine how the exact envelope is read from (z, θ) : the deterministic positive-mode control, the support E , the maximizing-owner frame used by the current finite macro system, and the normalized profile θ when that profile belongs to a fixed finite palette.

Definition 5.12 (Finite exact projective mode system). A positive factor lies in a finite exact projective mode system $\mathcal{P}$ if there is a finite set V of modes and fixed vectors θ_v such that at every cut in mode v ,

$$H = z\mathbf{1} + \theta_v$$

on the recorded support, and every macro transition of the factor updates the deterministic A height and the common scalar z additively while changing only the finite mode.

The phrase *exact* means that support, threshold regime, and the affine readout of every finite coordinate are part of the mode; no selected-run approximation is used.

Proposition 5.13 (Finite-projective compression). *Fix a concrete exact entry, a finite exact projective system $\mathcal{P}$, and one exact deterministic target configuration. There is an effective bound on every shortest factor staying in $\mathcal{P}$ and reaching that target. Moreover one can compute an endpoint-dominating shortest representative.*

Proof. Use deterministic height and the common scalar z as two ordinary VASS counters; the projective mode is finite control. Reachability sets of two-dimensional VASS with states are effectively semilinear [4]; hence, after fixing the target mode and exact deterministic target height, the one-dimensional set of reachable target scalars z is effectively semilinear and its emptiness and least element are decidable. Let $z_{\min}$ be that least element whenever the target slice is nonempty. Exhaustive word enumeration then computes a shortest word reaching (target mode, target A configuration, $z_{\min}$). Any observed factor exiting with scalar $z \geq z_{\min}$ is endpoint-dominated by this representative. Taking the maximum over the finitely many nonempty target modes gives the bound; empty target slices contribute nothing. $\square$

Remark 5.14 (Safe use of alternative paths). Proposition 5.13 is exactly the place where an alternative VASS path is legitimate: it is used to *construct a replacement word* with the same deterministic endpoint and a smaller exact envelope. No claim is made that the observed literal word itself belongs to a particular VASS path scheme.

5.4 Owner guards, recurrent guard rank, and the rank-one base

After finite support/provenance and maximizing-owner refinement, the remaining owner changes are governed by finitely many affine comparisons

$$g_j(x) = \ell_j(x) - k_j,$$

where x denotes the projective owner/resource vector. Fix one recurrent finite owner/chamber strongly connected component S . Let W_S be the rational span of projective displacements of directed cycles of S , after common translation is factored out.

Definition 5.15 (Recurrent guard rank). The recurrent guard rank of S is

$$\rho(S) = \dim_{\mathbb{Q}} \text{span}\{\ell_j|_{W_S} : j \text{ switches recurrently in } S\}.$$

Only restrictions to the recurrent cycle space count. Many syntactic guards may therefore have rank one.

The base of the induction is stronger than mere compactness.

Lemma 5.16 (Common scalar for rank-one guards). *If $\rho(S) = 1$, there is an effectively computable integer linear functional μ on the cycle lattice and integers a_j such that*

$$\ell_j(w) = a_j \mu(w)$$

for every recurrent cycle displacement $w \in W_S$.

Proof. The restricted functionals span a one-dimensional rational subspace of W_S^* . Choose a rational generator and clear denominators simultaneously. $\square$

For an edge $e : u \rightarrow v$ of the finite owner SCC with translation c_e , the residual

$$r_j(e) = \ell_j(c_e) - a_j \mu(c_e)$$

has zero sum on every directed cycle. Hence it is a finite-control coboundary: $r_j(e) = \phi_j(v) - \phi_j(u)$. Along a path from a fixed entry, with

$$T = \mu(x - x_0),$$

every recurrent guard therefore has the form

$$g_j(x) = a_j T + b_{j,v} \tag{3}$$

for a finite table of constants $b_{j,v}$.

Proposition 5.17 (Rank-one compilation). *Every recurrent owner SCC of guard rank at most one can be compiled effectively into a VASS with one zero-tested counter. Consequently, from a concrete entry and exact deterministic target configuration, exact target-slice nonemptiness and finite Pareto-minimal endpoint bases are computable; in particular the SCC is shortest-witness compact.*

Proof. For rank zero all recurrent guard truths are fixed after finite refinement and their slacks are ordinary counters. For rank one, the finitely many affine thresholds in (3) partition $\mathbb{Z}$ into two unbounded rays and finitely many bounded cells. On an unbounded ray store the distance of T from its boundary in one distinguished counter s ; bounded cells store the exact value of T in finite control. A macro edge changes T by a fixed integer. A cell crossing can occur only when s belongs to a computably bounded source band. Drain s through a bounded finite-control chain, use the unique zero test to verify the crossing value, remember the consumed bounded amount in finite control, and reconstruct the target slack. All other resources are ordinary VASS counters. Reachability for VASS with one zero test is decidable. An endpoint slice fixing some counters exactly and leaving the others free reduces to one exact-configuration query by entering a private cleanup phase that decrements only the free counters and leaves the fixed counters unchanged; the queried cleanup target fixes the prescribed counters and sets every formerly free counter to zero. Hence exact endpoint-slice nonemptiness is decidable from the same oracle.

Effective Pareto algorithm. The basis computation uses exact-slice reachability directly and does not rely on detecting when a growing product box is large enough. More generally, let R_F be the reachable endpoint set after fixing some owner coordinates F exactly. First decide whether R_F is empty. If it is nonempty, enumerate one reachable endpoint $a \in R_F$. Every componentwise-minimal $x \in R_F$ satisfies either $x = a$ or $x_i < a_i$ for at least one still-free coordinate i . For every such i and every integer $0 \leq \zeta < a_i$, recursively solve the lower-dimensional exact slice $R_{F \cup \{x_i = \zeta\}}$. Each recursive call fixes one additional coordinate and each branching range is finite, so the recursion terminates. Taking the union of the returned candidates with a and removing dominated vectors gives exactly $\text{Min}(R_F)$. At the root this computes the full finite Pareto basis. For each minimum, breadth-first word enumeration yields a shortest representative. Thus effectivity follows from the one-zero-test exact-slice oracle, not from Dickson finiteness alone.

Since the exact envelope is an affine readout of the owner resources in the fixed refined mode, a Pareto-minimal representative endpoint-dominates any longer competitor and Lemma 5.5 applies. $\square$

5.5 Directional walls and exact finite wall signatures

Assume now $\rho \geq 2$. Choose a basis $\mu_1, \dots, \mu_\rho$ for the recurrent guard directions. A guard is μ_1 -sensitive if its restriction has a nonzero μ_1 coefficient. A μ_1 wall event is a transition at which the truth value of at least one μ_1 -sensitive guard changes.

Every such switch occurs in a finite source band: the guard changes by a fixed integer on a fixed macro edge, so its source value must lie between two consecutive bounded thresholds.

Definition 5.18 (Exact wall signature). An exact μ_1 -wall signature records

- (a) the source and target refined finite modes and deterministic controls;
- (b) the set of μ_1 -sensitive guards that change truth and their orientations;
- (c) the exact bounded source guard/slack values at the crossing;
- (d) the current support/provenance color and owner frame;
- (e) the crossing macro edge.

The set $\mathcal{W}$ of exact wall signatures is finite and effectively enumerable.

Finiteness follows because all control/frame/support components are finite and every switching guard source value belongs to an edge-dependent finite threshold band.

Lemma 5.19 (Wall-free rank descent). *Every maximal μ_1 -wall-free strip has recurrent guard rank at most $\rho - 1$.*

Proof. Every μ_1 -sensitive guard has fixed truth throughout the strip and is carried as an ordinary nonnegative slack. Only guards whose μ_1 coefficient is zero can switch, and their restrictions lie in $\text{span}(\mu_2, \dots, \mu_\rho)$. $\square$

A bounded number of walls can also be handled without creating a second zero test.

Proposition 5.20 (Bounded-wall compilation). *Fix K . A rank- ρ factor containing at most K μ_1 -wall events yields either a responsible rank- $\leq \rho - 1$ escape or an effective endpoint-dominating length bound, assuming the lower-rank compactness invariant.*

Proof. The factor is a concatenation of at most $K + 1$ wall-free strips and K crossing macros. If normalized/projective/anchored/common-shift data escape in a strip, Lemma 5.19 localizes a genuine lower-rank escape. Otherwise every strip belongs to a finite lower-rank bounded-context palette and is compact by induction; the crossing macros are finite. Summing the finitely many bounds gives the factor bound. If one wants an exact compilation rather than merely an extensional replacement, an exact bounded wall value $s = c$ can be checked without a fresh zero test by subtracting c , retiring the old ordinary slack counter, and requiring it to be zero only at the final exact-reachability target. This is needed only finitely many times because K is fixed. $\square$

Hence a genuinely rank- ρ obstruction must have unbounded wall multiplicity.

5.6 Dense same-signature atoms and carrier colors

Let $B = |\mathcal{W}|$. Write the wall-signature word of a factor as $w_1 \cdots w_N$.

Lemma 5.21 (Dense same-signature returns). *If $N \geq 4B$, some signature $\sigma \in \mathcal{W}$ has at least*

$$\left\lfloor \frac{N}{4B} \right\rfloor - 1$$

pairwise interior-disjoint consecutive σ -to- σ return intervals, each containing at most

$$K_{\text{dense}} = 4B + 1$$

wall events.

Proof. Some σ occurs at least $\lceil N/B \rceil$ times. The sum of the consecutive occurrence gaps is $< N$. If more than half of these gaps exceeded $4B$, their sum would exceed N . The stated coarse linear lower bound follows. $\square$

Returning to the same exact signature fixes one independent μ_1 -sensitive wall equation. Thus the endpoint displacement of every dense atom lies in $W_S \cap \ker \ell_\sigma$, and its endpoint recurrent guard rank is at most $\rho - 1$.

Definition 5.22 (Carrier color). For a dense atom, a left or right *carrier* is the bounded neighborhood from the exact wall endpoint to the canonical shell/interface cut used for comparison. Under a fixed bounded-context cutoff, its carrier color records the exact wall signature, side (left/right), deterministic control, support/provenance color, owner frame, bounded vertical offset from the shell level, and all normalized envelope/guard data in the prescribed finite cutoff box.

Carrier colors are therefore finite whenever the context is bounded. If the boundary-to-canonical correction leaves this finite palette, the first escaping quantity occurs in one of finitely many wall-free strip positions inside a bounded-wall carrier, and Lemma 5.19 localizes the escape below rank ρ .

5.7 High walls and translated arches

If the heights of all recurrent walls are bounded, the exact physical wall contexts belong to a finite set and Proposition 5.20 closes the branch. Thus the residual case has arbitrarily high wall centers. Around a wall center t of height a , for each $1 \leq \ell < a$ define the last/first height- ℓ cuts

$$u_\ell^t < t < v_\ell^t.$$

Their intervening factors are nested positive arches. Combining the dense atom skeleton, finite carrier colors, and ordinary finite Ramsey refinement yields an arbitrarily deep family

$$J_\ell = e_\ell^L \longrightarrow e_\ell^R$$

with fixed left and right carrier colors and fixed offsets

$$d(e_\ell^L) = \ell + r_L, \quad d(e_\ell^R) = \ell + r_R.$$

The carriers have recurrent rank at most $\rho - 1$.

The next lemma makes the finite-color argument explicit.

Lemma 5.23 (Translated-arch localization). *Let $J_{h_0}, \dots, J_{h_m}$ be a sufficiently long fixed-carrier translated-arch family with strictly increasing levels. Then either*

- (i) *two levels satisfy the anchored deletion inequalities of Lemma 5.6; or*
- (ii) *after passing to a homogeneous subfamily, one fixed side $X \in \{L, R\}$ and one fixed live coordinate $q \in Q_N$ satisfy a strict chronological chain*

$$F_{x_0}(q) > F_{x_1}(q) > \dots > F_{x_n}(q),$$

where x_i are the selected canonical endpoints/carrier cuts on side X .

In case (ii), the escape is either carried inside one lower-rank carrier strip or is visible on the same-signature lower-rank endpoint system.

Proof. For each pair of levels $h < k$, first ask whether both anchored comparisons required by Lemma 5.6 hold. If yes, case (i) occurs. Otherwise color the pair by the first failed side (left or right) and the least envelope coordinate witnessing that failure. Support and side are already fixed, so this is a finite coloring by at most $2|Q_N|$ colors. Ramsey's theorem gives a long homogeneous set. If the homogeneous color is, for example, “left, q ”, then for every earlier/later pair the left comparison fails in the same orientation, hence integrality and total order on the selected coordinate give a strict monotone chain after reversing orientation if necessary. The right-side case is symmetric.

Now compare each canonical endpoint with the corresponding exact wall endpoint through its fixed carrier. There are finitely many strip positions. If the correction in coordinate q or any other normalized coordinate is unbounded, one fixed wall-free strip position carries unbounded variation and has rank at most $\rho - 1$. If all corrections are bounded, pass to a finite refinement fixing them; the strict drift then transfers to the exact same-signature wall endpoints. Their endpoint displacement lies in the wall kernel, so their recurrent endpoint rank is at most $\rho - 1$. $\square$

Absolute shell height is not itself a new resource on a fixed carrier: once the vertical range about level ℓ is bounded and all zero tests are inactive, translating the whole carrier upward preserves its transition structure. Hence only the normalized/projective, anchored-relative, common-shift, and relative A -transfer quantities can escape.

5.8 Finite scalar/projective reserve split and the vector shell defect

Lemma 5.23 leaves a deletion-free translated-arch family in which one endpoint side carries an arbitrarily long strict anchored drift. The final proof uses a finite reserve split; no infinitary Dickson extraction on normalized profiles is used.

Assume for definiteness that the responsible side is the left side. At the selected left endpoints write

$$F_\ell^L = H_{e_\ell^L} - d(e_\ell^L)\mathbf{1}.$$

After stable-support refinement, finite pigeonhole fixes one live coordinate r attaining the minimum of F_ℓ^L on an arbitrarily long subfamily. Put

$$a_\ell := F_\ell^L(r), \quad \theta_\ell := F_\ell^L - a_\ell \mathbf{1}. \quad (4)$$

Then $\theta_\ell \in \mathbb{N}^S$, $\theta_\ell(r) = 0$, and $F_\ell^L = a_\ell \mathbf{1} + \theta_\ell$. By Lemma 5.23, on the selected chain one fixed coordinate q satisfies

$$F_{h_0}^L(q) > F_{h_1}^L(q) > \cdots > F_{h_n}^L(q).$$

Lemma 5.24 (Effective projective cutoff). *For the concrete first-return entry and fixed finite support/regime palette there is a computable constant C_{proj} with the following property. If a normalized endpoint profile has*

$$\max_{p \in S} \theta(p) > C_{\text{proj}},$$

then a bounded nonzero N/N differential core is exposed on a fixed refined route, unless a provenance-rank drop occurs first. The branch therefore transfers to the projective/lower-rank localization machinery. In the complementary branch every normalized profile lies in the finite box $[0, C_{\text{proj}}]^S$.

Proof. Track a fixed difference $H(p) - H(r)$ in the finite synchronized owner/product graph. After cycle erasure, a simple residual has effectively bounded length and hence bounded differential effect. If the difference exceeds the concrete entry spread plus twice the maximal simple-residual contribution (enlarged over the finite support/regime event palette), some primitive recurrent cycle has nonzero differential effect. Finite refinement fixes its sign and route type. If the responsible owner/provenance data cannot be stabilized before that extraction, the provenance branch has already descended. Taking the maximum of these finitely many simple-path/event bounds gives C_{proj} . $\square$

Lemma 5.25 (Finite scalar/projective reserve split). *For every n -step strict endpoint chain as above,*

$$(a_{h_0} - a_{h_n}) + (\theta_{h_0}(q) - \theta_{h_n}(q)) \geq n. \quad (5)$$

Consequently either the chain crosses the projective cutoff of Lemma 5.24, or finite pigeonhole extracts an arbitrarily long subchain on which one exact profile

$$\theta_\ell = \bar{\theta}$$

is fixed and the scalar sequence a_ℓ decreases strictly.

Proof. Integrality gives $F_{h_0}^L(q) - F_{h_n}^L(q) \geq n$. Substitute $F^L(q) = a + \theta(q)$ to obtain (5). If some profile leaves the finite projective box, Lemma 5.24 transfers the branch. Otherwise only finitely many exact profiles occur, so finite pigeonhole—not Dickson’s lemma—gives an arbitrarily long constant-profile subchain. Strict decrease of $F^L(q)$ then forces strict decrease of a . $\square$

From now on we work in this canonical no-projective-escape branch and write $\theta_\ell = \bar{\theta}$. Take $h < k$ in the fixed-profile subchain and put

$$c := k - h, \quad \delta := a_h - a_k > 0.$$

The fixed endpoint offsets give the exact left-envelope identity

$$H_k^L = H_h^L + (c - \delta)\mathbf{1}. \quad (6)$$

Lemma 5.26 (Translated input domination). *If $\delta \leq c$ and H' is the exact net envelope obtained by reading the literal inner word J_k from the outer left envelope H_h^L , then*

$$H' \leq H_k^R - (c - \delta)\mathbf{1}. \quad (7)$$

Proof. Equation (6) gives

$$H_h^L + (c - \delta)\mathbf{1} = H_k^L.$$

For a one-counter net, common input credit is monotone under a fixed word: for every $t \geq 0$,

$$T_x(H + t\mathbf{1}) \geq T_x(H) + t\mathbf{1}.$$

Applying this to $x = J_k$ gives

$$H' + (c - \delta)\mathbf{1} \leq T_{J_k}(H_k^L) = H_k^R.$$

□

Define the *vector shell defect*

$$Y_\ell := F_\ell^R - a_\ell\mathbf{1}, \quad F_\ell^R := H_\ell^R - d(e_\ell^R)\mathbf{1}. \quad (8)$$

Because $d(e_k^R) - d(e_h^R) = c$,

$$H_k^R - (c - \delta)\mathbf{1} = H_h^R + (Y_k - Y_h). \quad (9)$$

Lemma 5.27 (Vector shell-defect pinning). *Let $h < k$ belong to the fixed-profile subchain and assume $\delta \leq c$. Exactly one of the following sufficient outcomes occurs.*

- (i) $Y_k \leq Y_h$ componentwise, in which case replacing the outer arch by the translated inner arch gives an endpoint-dominating strict shortening;
- (ii) there is a live coordinate s such that

$$1 \leq R_h^R(s) - H_h^R(s) \leq Y_k(s) - Y_h(s). \quad (10)$$

Proof. Combine Lemma 5.26 with (9). If $Y_k \leq Y_h$, the translated exit envelope is at most H_h^R , so Lemma 5.5 shortens the shortest witness. Otherwise the translated word is still accepted by A and is strictly shorter, hence R -minimality forces its exit envelope to satisfy the requirement. At the outer right cut the original word has $H_h^R < R_h^R$, while the translated exit reaches the suffix requirement in some coordinate s , giving (10). □

A pair is *nonsuperlevel* when $\delta \leq c$ and *superlevel* when $\delta > c$. Finite Ramsey refinement yields a long homogeneous subfamily of one kind. In the nonsuperlevel case, Lemma 5.27 and a finite coloring by the least positive coordinate of $Y_k - Y_h$ give a fixed coordinate with strict Y -growth. The superlevel case is threshold-sensitive and is developed next.

5.9 Superlevel activation and the vector-defect/activation split

Fix a superlevel pair $h < k$ and put

$$d := \delta - c > 0. \quad (11)$$

Then (6) becomes

$$H_k^L = H_h^L - d\mathbf{1}, \quad (12)$$

and the corresponding right identity is

$$H_k^R = H_h^R - d\mathbf{1} + (Y_k - Y_h). \quad (13)$$

Suppose first that $Y_k \leq Y_h$. Then the inner right envelope is strictly dominated by the outer right envelope. Let

$$Q_{h,k} := S_{J_k}(R_h^R) \quad (14)$$

be the exact entry requirement for reading the literal inner word J_k and then the inherited outer suffix.

Lemma 5.28 (Exact superlevel activation certificate). *If $h < k$ is superlevel and $Y_k \leq Y_h$, then*

$$H_k^L < Q_{h,k}$$

componentwise, while shortestness implies $H_h^L \not\leq Q_{h,k}$. Hence there is a live coordinate p such that

$$H_k^L(p) < Q_{h,k}(p) \leq H_h^L(p). \quad (15)$$

Writing

$$\eta := Q_{h,k}(p) - H_k^L(p),$$

one has the exact bound

$$1 \leq \eta \leq d. \quad (16)$$

Proof. Since $Y_k \leq Y_h$, equation (13) gives $H_k^R < H_h^R < R_h^R$. Exact residuation, Theorem 3.4, yields $H_k^L < S_{J_k}(R_h^R) = Q_{h,k}$. Replacing J_h by the downward translation of J_k is a legal strict deterministic shortening. Hence R -minimality forces the shortened replacement to meet the inherited requirement, equivalently $H_h^L \not\leq Q_{h,k}$. Choose p satisfying (15). Since $H_h^L(p) - H_k^L(p) = d$ by (12), the gap satisfies (16). $\square$

This is an exact threshold statement: the actual inner input is short of the virtual continuation threshold by η , while the outer input crosses that threshold. It is not yet a statement about a forward maximizing strand of the original rejected word.

For every sufficiently separated superlevel pair color the pair by

- A if $Y_k \leq Y_h$ componentwise;
- otherwise V_s , where s is the least coordinate with $Y_k(s) > Y_h(s)$.

There are finitely many colors.

Lemma 5.29 (Activation-or-vector-defect dichotomy). *Every arbitrarily deep superlevel tower contains arbitrarily long subfamilies of exactly one of the following forms.*

- (i) Activation-critical: *for every earlier/later pair Y is componentwise nonincreasing and Lemma 5.28 supplies an exact threshold gap η ;*
- (ii) Vector-defect: *one fixed coordinate s satisfies*

$$Y_{h_0}(s) < Y_{h_1}(s) < \dots$$

Proof. Apply finite Ramsey to the coloring above. Homogeneous color A gives the first alternative and homogeneous color V_s gives the second. $\square$

The vector-defect alternative is not left abstract. Fix a live right coordinate s and, for each ℓ , choose one maximizing N -path $\pi_\ell : p_\ell \rightarrow s$ over the literal arch J_ℓ realizing $H_\ell^R(s)$. Pass to a subsequence with fixed input state p . Put

$$\kappa_A := r_R - r_L = \Delta_A(J_\ell).$$

Because the left projective profile is fixed to $\bar{\theta}$, the exact owner/path identity is

$$Y_\ell(s) = \bar{\theta}(p) + \omega_\ell, \quad \omega_\ell := \Delta_N(\pi_\ell) - \kappa_A. \quad (17)$$

Lemma 5.30 (Actual maximizing-strand source of vector defect). *If $Y_\ell(s)$ is unbounded above in the canonical no-projective-escape branch, then $\omega_\ell \rightarrow \infty$ along a subsequence. Cycle erasure on the actual maximizing synchronized paths exposes a fixed primitive synchronized cycle with positive relative effect $\Delta_N - \Delta_A > 0$, unless a provenance-rank drop occurs first.*

Proof. The term $\bar{\theta}(p)$ in (17) is fixed, so unbounded Y is exactly unbounded positive relative gain on the chosen literal maximizing strands. Cycle erasure in the finite synchronized positive-mode product leaves a bounded simple residual. Hence recurrent primitive cycles carry unbounded positive relative contribution, and finite pigeonhole fixes one positive type unless the owner/provenance refinement changes first. $\square$

Thus only the activation-critical branch is new.

5.10 Literal activation annuli and fixed affine cells

Let $h < k$ be an activation-critical pair. The outer arch factors literally as

$$J_h = L_{h,k} J_k R_{k,h},$$

where $L_{h,k}$ runs from e_h^L to e_k^L and $R_{k,h}$ from e_k^R to e_h^R . The deterministic effects are $+c$ and $-c$, respectively.

Lemma 5.31 (Matched anchored drift in an activation annulus). *For every activation-critical pair,*

$$F_k^L(r) - F_h^L(r) = -\delta,$$

and, in chronological order on the right flank,

$$F_h^R - F_k^R = (Y_h - Y_k) + \delta \mathbf{1} \geq \delta \mathbf{1}.$$

Equivalently, in absolute envelopes,

$$H_h^L(r) - H_k^L(r) = d, \quad H_h^R - H_k^R \geq d \mathbf{1}. \quad (18)$$

Proof. The left identity follows from $F_\ell^L(r) = a_\ell$. The right identity follows from $F_\ell^R = Y_\ell + a_\ell \mathbf{1}$ and $Y_h \geq Y_k$. Restoring the deterministic height difference c converts δ into $d = \delta - c$ in the absolute envelopes. $\square$

The two literal flanks of one annulus therefore carry a matched credit valley: the left loses anchored mass and the right restores at least the corresponding common amount. If adjacent cells have unbounded values of δ or unbounded endpoint increments, the responsible mass is already localized to one of finitely many bounded-wall flank positions and hence to lower rank. We may therefore restrict to bounded cells.

After finite refinement fix integers

$$c, \delta, d = \delta - c > 0, \quad p, \quad 1 \leq \eta \leq d$$

and define the right adjacent-cell increment

$$v_i := Y_i - Y_{i+1} \geq 0.$$

The left profile is already fixed by Lemma 5.25, so the endpoint recurrences are

$$H_{i+1}^L = H_i^L - d\mathbf{1}, \quad H_i^R = H_{i+1}^R + d\mathbf{1} + v_i. \quad (19)$$

Lemma 5.32 (Finite affine-cell refinement). *If some coordinate of v_i is unbounded, a bounded-height annulus carries unbounded normalized right-endpoint change and hence localizes to a lower-rank projective/relative escape or a provenance-rank drop. Otherwise, after finite refinement, $v_i = v$ is one fixed vector for every recurrent cell.*

Proof. A cell has fixed shell width c and fixed wall/carrier types, hence only finitely many wall-free strip positions. Unbounded normalized endpoint change forces one fixed position to carry unbounded change; Lemma 5.19 lowers the recurrent rank there. In the no-escape case v_i ranges over a finite box and may be fixed by finite refinement. $\square$

Fixed endpoint affine data do not imply that the cell interiors form one lower-rank machine. We use only the endpoint recurrences and the extensional replacement principle.

5.11 Pure-scalar recurrence and the virtual threshold shadow

The terminal fixed-cell survivor simplifies further.

Lemma 5.33 (Scalarization of stable cells). *In a recurrent fixed affine cell family with no endpoint-visible projective escape or provenance-rank drop, one has*

$$v = \lambda \mathbf{1}$$

for some fixed $\lambda \geq 0$. Hence

$$H_{i+1}^L = H_i^L - d\mathbf{1}, \quad H_{i+1}^R = H_i^R - (d + \lambda)\mathbf{1}. \quad (20)$$

Proof. The left normalized profile is already fixed. If two coordinates of v differ, repeated cells produce linearly unbounded right projective spread, contradicting the no-projective-escape hypothesis. Hence v is a common scalar vector. $\square$

Let

$$Q_i := S_{J_{i+1}}(R_i^R)$$

be the replacement-entry requirement of cell i , and retain the fixed activation coordinate p with

$$Q_i(p) - H_{i+1}^L(p) = \eta.$$

Equation (20) gives the exact threshold ladder

$$Q_{i+1}(p) = Q_i(p) - d, \quad H_i^L(p) - Q_i(p) = d - \eta. \quad (21)$$

Put $K := d - \eta \geq 0$ and $q_i := Q_i(p)$. Then

$$H_{i+1}^L(p) = q_i - \eta, \quad H_i^L(p) = q_i + K. \quad (22)$$

For each i choose canonically an N -path $\pi_i : p \rightarrow s_i$ over J_{i+1} attaining the minimum in the backward transformer at coordinate p :

$$q_i = \max\{\text{req}(\pi_i), R_i^R(s_i) - \text{eff}(\pi_i)\}. \quad (23)$$

After finite refinement the exit state $s_i = s$ is fixed.

There are two formal ways for the maximum in (23) to equal q_i . If $\text{req}(\pi_i) = q_i$, the threshold run reaches counter 0 at an internal prefix. If instead the terminal term equals q_i and $\text{req}(\pi_i) \leq q_i - 1$, minimality forces exact tightness along the path.

Lemma 5.34 (Virtual tight-shadow principle). *In the terminal-tight case, let $C_i(x)$ be the counter value of the threshold run on π_i at a cut x , and let $B_i(x)$ be the exact backward requirement at that path state for the remaining virtual continuation J_{i+1} followed by the inherited outer suffix. Then*

$$B_i(x) = C_i(x) \quad (24)$$

for every cut x of π_i .

Proof. The chosen suffix is feasible from $C_i(x)$, so $B_i(x) \leq C_i(x)$. If $B_i(x) \leq C_i(x) - 1$, then because the whole selected path has internal requirement at most $q_i - 1$, starting the prefix one unit lower remains feasible up to x , and the exact continuation is feasible from the resulting $C_i(x) - 1$. This contradicts the minimality of $q_i = Q_i(p)$. $\square$

Thus the accepting translated replacement, which starts K units above threshold, stays exactly K above the virtual backward requirement along a terminal-tight path.

5.12 Collapse of the tight shadow to a bounded high contact

The tight-shadow alternative is not a separate terminal branch.

Lemma 5.35 (Terminal tightness forces a contact). *In the terminal-tight case,*

$$q_i - \eta < \text{req}(\pi_i) \leq q_i - 1.$$

Consequently some prefix of π_i attains threshold counter

$$j_i := q_i - \text{req}(\pi_i) \in \{1, \dots, \eta - 1\}.$$

At that cut the exact virtual backward requirement equals j_i , while the actual translated replacement path has counter $K + j_i$.

Proof. Assume instead that $\text{req}(\pi_i) \leq q_i - \eta = H_{i+1}^L(p)$. Then the selected path is internally feasible from the actual inner envelope coordinate. Its endpoint counter is $R_i^R(s) - \eta$. But the pure-scalar right recurrence raises the outer right envelope over the inner one by $d + \lambda$, and $\eta \leq d$, giving $H_i^R(s) \geq R_i^R(s)$, contradicting rejection at the outer right cut. The rest follows from Lemma 5.34. $\square$

In the internal-contact case the threshold run itself reaches counter 0, so the virtual requirement at that cut is 0 and the actual replacement counter is exactly K . Combining both cases yields the following uniform normal form.

Theorem 5.36 (Bounded virtual contact normal form). *After finite refinement, every recurrent terminal activation cell contains a cut on a concrete accepting translated-replacement path with*

- one fixed N contact state t ;
- one fixed virtual backward requirement $b \in \{0, \dots, \eta - 1\}$ at t ;

- one fixed actual replacement counter $b + K$.

Moreover the deterministic A height at these contacts tends to infinity along every arbitrarily deep translated-arch tower.

Proof. The first three items follow from finite refinement of the two contact cases above. High-shell geometry gives a constant M_* such that every translated inner arch in the fixed carrier family stays at height at least its shell level minus M_* . The shell levels tend to infinity, so the contact height does as well. $\square$

Let B_i be the full exact backward-requirement vector at the virtual contact. Since $B_i(t) = b$ is fixed, either some normalized difference $B_i(q) - B_i(t)$ escapes, in which case the backward differential argument localizes a lower-rank/projective escape, or the complete vectors B_i lie in a finite set and may be fixed to one exact vector B_* . In the latter branch the contact has fixed finite N context but arbitrarily high deterministic height.

5.13 From bounded contacts to an actual-trace relative valley

The threshold ladder (21) implies that in a tower of $n + 1$ recurrent cells the outer threshold is at least nd . Thus along a selected outer sequence

$$q_i \longrightarrow \infty. \quad (25)$$

Fix the bounded contact normal form of Theorem 5.36. Let P_i be the literal prefix of the translated inner arch from its translated start to the contact. On the chosen accepting replacement path, the N counter starts at $q_i + K$ and reaches $b + K$, so

$$\Delta_N(P_i) = b - q_i. \quad (26)$$

Let $\alpha_i := \Delta_A(P_i)$. High-shell geometry bounds α_i from below by a constant independent of i . Therefore

$$\omega(P_i) := \Delta_N(P_i) - \Delta_A(P_i) \longrightarrow -\infty. \quad (27)$$

Lemma 5.37 (Contact-prefix negative relative mass). *Every arbitrarily deep terminal activation tower contains literal synchronized prefixes P_i of the translated/original inner-arch words with $\omega(P_i) \rightarrow -\infty$. Cycle erasure on these actual synchronized paths exposes a fixed primitive cycle type with negative relative effect, unless the responsible mass has already localized to a lower-rank wall-free strip or support/provenance drop.*

Proof. Equation (27) is exact. The synchronized positive-mode product is finite. Removing all cycle occurrences leaves a simple path of uniformly bounded relative effect, so unbounded negative relative mass must be carried by recurrent primitive cycles. Finite pigeonhole fixes one responsible type. If its occurrences have unbounded internal wall-free variation, Lemma 5.19 already gives the lower-rank alternative; otherwise bounded-wall packet refinement applies. $\square$

Let T_i be the literal suffix of the same concrete accepting translated-replacement run from the bounded contact to the zero-return cut z terminating the positive first-return corridor. At the contact the deterministic counter is $D_i \rightarrow \infty$ and the selected N counter is the fixed value $b + K$. At z the deterministic counter is 0, while the selected accepting N run has some counter $c_i \geq 0$. Hence

$$\Delta_A(T_i) = -D_i, \quad \Delta_N(T_i) = c_i - (b + K) \geq -(b + K),$$

and therefore

$$\omega(T_i) = \Delta_N(T_i) - \Delta_A(T_i) \geq D_i - (b + K) \longrightarrow +\infty. \quad (28)$$

This is the canonical post-contact argument: no split according to the A -transfer to the contact is needed.

Theorem 5.38 (Relative valley). *After excluding strict lower-rank/projective/support escapes, every terminal activation tower contains, on one concrete accepting translated-replacement run and in chronological order,*

$$P_i \longrightarrow (t, b + K; A = D_i) \longrightarrow T_i$$

with a fixed bounded N contact, $D_i \rightarrow \infty$, and

$$\omega(P_i) \rightarrow -\infty, \quad \omega(T_i) \rightarrow +\infty.$$

The suffix T_i ends at the actual zero-return cut of the positive first-return corridor. All factors are literal factors of the same accepting translated-replacement word; no alternative VASS path is used.

Proof. The negative side is Lemma 5.37. The positive side is the exact zero-return calculation above. The fixed contact state/counter and diverging deterministic height are supplied by Theorem 5.36. No path-scheme or fixed-transfer assumption enters the argument. $\square$

Theorem 5.38 is the only input needed for the actual-trace cycle analysis below. Importantly, it does not say that the observed trace belongs to a fixed path scheme.

5.14 Trace fidelity: actual cycles versus alternative paths

The activation branch needs a second distinction in addition to the projective-mode and recurrent-guard-rank definitions. A generic 2-VASS flattening theorem says that if one synchronized configuration reaches another, some alternative path belonging to a finite family of path schemes reaches it. It does *not* imply that the observed literal input factor itself belongs to one of those schemes. Therefore path-scheme periods may not be deleted from the observed word unless their literal occurrences have been independently located.

We use the following two rules throughout.

- (T1) **Observed-word operation.** Any literal deletion package is extracted by cycle erasure on the actual synchronized trace. Its blocks occur at identified positions of the original word.
- (T2) **Replacement-word operation.** VASS reachability or flattening may construct a different word only when Lemma 5.5 is invoked at the common deterministic endpoint, as in Proposition 5.13.

Consider now an actual synchronized positive-mode trace before a bounded contact. For a simple synchronized cycle C put

$$v(C) = (\Delta_N(C), \omega(C)), \quad \omega(C) = \Delta_N(C) - \Delta_A(C).$$

Cycle-erasing the actual trace leaves a bounded simple residual and a multiset of primitive simple cycle occurrences. Let $C_1, \dots, C_M$ be the finite simple-cycle types and $x_{i,j}$ their multiplicities in the i th actual pre-contact path. Pass to a subsequence so that, for each j , either $x_{i,j}$ is bounded or $x_{i,j} \rightarrow \infty$. Let I_∞ be the latter types and put

$$K_\infty = \text{cone}_{\mathbb{Q}_{\geq 0}} \{v(C_j) : j \in I_\infty\}.$$

Lemma 5.39 (Actual-trace unbounded-multiplicity cone). *If $\Delta_N(P_i) \rightarrow -\infty$ and $\omega(P_i) \rightarrow -\infty$ on the actual pre-contact paths, then K_∞ meets the open southwest quadrant*

$$\text{SW} = \{(x, y) : x < 0, y < 0\}.$$

Every generator used in K_∞ occurs with unbounded actual multiplicity in the observed literal traces.

Proof. The bounded-multiplicity cycle types and the simple residual contribute only $O(1)$. If K_∞ were disjoint from the open southwest quadrant, the rational polyhedral separation theorem would give a nonzero nonnegative linear functional that is nonnegative on K_∞ . Applied to the cycle decomposition, that functional would bound from below the corresponding nonnegative combination of $(\Delta_N(P_i), \omega(P_i))$, contradicting that both coordinates tend to $-\infty$. The last statement is the definition of I_∞ . $\square$

Lemma 5.40 (Support-two southwest reduction). *If a finitely generated rational cone in $\mathbb{R}^2$ meets SW, it contains a southwest vector supported on at most two generators. If the support is two and neither generator is southwest, then after relabeling the two actual cycle types S, D satisfy*

$$\begin{array}{c|ccc} & \Delta_A & \Delta_N & \omega \\ \hline S & > 0 & \geq 0 & < 0 \\ D & < 0 & < 0 & \geq 0. \end{array}$$

Proof. Apply the conic form of Carathéodory's theorem in dimension two to the cone K_∞ from Lemma 5.39. The selected one or two generators therefore belong to I_∞ and occur with unbounded actual multiplicity. For a support-minimal two-generator southwest combination, neither generator can have both coordinates negative. One must supply the negative N coordinate and the other the negative relative coordinate; substituting $\Delta_A = \Delta_N - \omega$ gives the stated opposite deterministic signs. $\square$

Write $a = \Delta_A(S) > 0$, $d = -\Delta_A(D) > 0$, $x = \Delta_N(S) \geq 0$, and $y = -\Delta_N(D) > 0$. Define

$$\rho = x/a, \quad \kappa = y/d.$$

A positive combination is southwest exactly when $\rho < \kappa$. With $g = \gcd(a, d)$ set

$$m = d/g, \quad n = a/g.$$

Then the primitive deterministic balance

$$\Pi = mS + nD$$

satisfies

$$\Delta_A(\Pi) = 0, \quad \Delta_N(\Pi) = \frac{ad}{g}(\rho - \kappa) < 0. \quad (29)$$

After clearing rational coefficients, m, n are fixed positive integers. By Lemma 5.39 and the proof of Lemma 5.40, both selected types lie in I_∞ ; sufficiently deep actual traces therefore contain the required finite number of literal occurrences of each type.

Proposition 5.41 (Deep trace-faithful neutral package deletion). *In a sufficiently high translated arch, delete from the original word the selected actual occurrences of one package Π , leaving the intervening connectors untouched. The modified deterministic run stays positive, rejoins the exact original A configuration after the last removed block, and strictly shortens the word.*

Proof. Each removed block is an actual A -control cycle in positive mode. The package contains a fixed number of fixed cycle types, so the maximum intermediate height discrepancy caused by deleting them in chronological order is bounded by a package constant. High-shell reserve eventually exceeds that constant. Equation (29) gives zero total deterministic effect, hence exact rejoining after the final deletion. $\square$

No statement about preserving an N run is made. Instead shortestness converts the A -safe deletion into a full-envelope certificate.

Lemma 5.42 (Critical deletion certificate). *Let H be the original exact envelope and H' the deleted-word envelope at the common splice after the last removed block; let R be the backward requirement of the unchanged suffix. Then for some live coordinate q ,*

$$H(q) < R(q) \leq H'(q).$$

Proof. The original factor is rejected against the fixed residual, so $H < R$. The deletion word is shorter and still accepted by A ; R -minimality therefore gives $H' \not\prec R$. The same coordinate witnesses both inequalities. $\square$

The singleton southwest case with $\Delta_A = 0$ is identical. Suppose instead that one actual primitive southwest cycle C occurs with unbounded multiplicity and $\Delta_A(C) \neq 0$. The literal translated endpoint arch containing these occurrences has one fixed total deterministic effect κ_A , independent of shell depth. Cycle-erasing that same actual synchronized arch leaves a simple residual of uniformly bounded deterministic effect and finitely many primitive cycle types. In the exact identity

$$\kappa_A = \text{res}_A(i) + \sum_j N_j(i) \Delta_A(C_j),$$

the contribution of C tends to one infinity sign. Therefore the sum of contributions of the opposite deterministic sign tends to the other; by finiteness one fixed actual primitive cycle type D of opposite A -effect also has unbounded multiplicity. With

$$g = \gcd(|\Delta_A(C)|, |\Delta_A(D)|), \quad m = |\Delta_A(D)|/g, \quad n = |\Delta_A(C)|/g,$$

the sparse literal package $mC + nD$ has total deterministic effect zero. Its cardinality is fixed, so high-shell reserve makes its chronological deletion A -safe and exact-rejoining. No sign condition on its net N effect is needed: shortestness alone yields Lemma 5.42. Thus the nonzero- A singleton folds into the same critical-deletion branch without any trace-changing path scheme.

5.15 Activation gain, threshold contacts, and noncircular return

At a critical deletion splice fix the decisive coordinate and write

$$\kappa_{\text{act}} = H'(q) - H(q) \geq 1.$$

There are two cases.

Bounded activation gain. If $\kappa_{\text{act}} \leq M$, then

$$1 \leq R(q) - H(q) \leq M.$$

The chronological rejection margin before the splice is therefore confined to a finite band. After support and normalized-profile refinement, every no-escape segment lies in finitely many exact diagonal/projective phases. Absorbing their finite interfaces produces one finite exact projective mode system, which closes by Proposition 5.13.

Unbounded activation gain. The deleted package contains only a fixed number of fixed literal blocks. Telescoping between successive deletion variants selects one fixed block whose removal causes unbounded later envelope sensitivity. Immediately after a fixed word C , two envelopes with stable support and bounded normalized entry data differ by at most an effective constant L_C : above the finite entry thresholds, every C -path is affine in the common scalar. Moreover, as long as two one-letter transformers enable the same transition family, max-plus evaluation is sup-norm nonexpansive. Therefore unbounded amplification must have a first step at which enabled-edge families differ, or support changes.

Lemma 5.43 (First-threshold contact). *Excluding an earlier support/projective escape, unbounded activation gain forces an aligned cut of the actual original shortest word, a fixed N state r , and an effective constant B_* such that*

$$H(r) \leq B_*.$$

The corresponding deterministic height tends to infinity along the high-shell family.

Proof. Choose the first suffix step where the previous uniform sup-distance bound fails. If support has not changed, some transition is enabled from the larger envelope and disabled from the smaller. Its fixed negative requirement pins the source coordinate of both envelopes into a bounded threshold band. There are only finitely many earlier deleted slots; bounded pairwise differences before the first responsible slot transfer the bounded coordinate back to the actual original witness by the triangle inequality. High-shell geometry supplies the diverging A height. $\square$

Thus at the contact

$$F(r) = H(r) - d \longrightarrow -\infty.$$

If the normalized N profile relative to r stays bounded, the contact belongs to a finite exact projective context. If it is unbounded, a genuine N/N projective difference escapes and localizes through a bounded-wall carrier to rank at most $\rho - 1$ or to a support/provenance drop.

Finally transfer the bounded contact to the canonical left endpoint of the translated arches. If that bounded correction itself escapes, it is already a lower-rank carrier escape. Otherwise after finite refinement one fixed reference coordinate has

$$H_\ell^L(r) = b, \quad d(e_\ell^L) = \ell + r_L.$$

Hence

$$a_\ell := F_\ell^L(r) = b - (\ell + r_L).$$

For $h < k$, with $c = k - h$,

$$a_h - a_k = c. \tag{30}$$

Lemma 5.44 (Contact-anchor noncircularity). *The superlevel activation condition $a_h - a_k > k - h$ cannot recur after a threshold contact is returned to the canonical endpoint system.*

Proof. Equation (30) gives equality for every pair. $\square$

This is the strict well-foundedness invariant missing from a naive recurring-wall argument.

5.16 Bounded-context domination and the rank lift

The induction does not need a global lower-rank relation or an arbitrary exact-slice oracle. It needs only the endpoint replacement guaranteed by Lemma 5.5.

Definition 5.45 (Bounded-context domination compactness, $\text{WCI}(\rho)$). Fix an actual positive-mode factor P in an R -minimal witness, one concrete exact entry, and one exact deterministic A exit. Assume its recurrent owner-guard rank is at most ρ , and assume that all normalized/projective, anchored-relative, common-shift, support/provenance, and deterministic transfer/range data relevant to the current carrier belong to fixed finite palettes. Then $\text{WCI}(\rho)$ asserts that there is an effective uniform bound on $|P|$; equivalently, every longer P has a strictly shorter endpoint replacement Q with the same exact deterministic exit and $H_Q \leq H_P$.

If one of the listed quantities leaves its finite palette, that first exit is not hidden inside WCI : it is declared a responsible escape and localized separately.

Proposition 5.17 proves $\text{WCI}(\rho)$ for $\rho \leq 1$.

Theorem 5.46 (Structural rank lift). *For every finite $\rho \geq 1$,*

$$\text{WCI}(\rho - 1) \implies \text{WCI}(\rho).$$

Proof. Fix a rank- ρ bounded-context factor.

Wall-free branch. Lemma 5.19 lowers the rank. If a tracked normalized quantity escapes, this is a responsible lower-rank escape; otherwise $\text{WCI}(\rho - 1)$ bounds the strip.

Bounded-wall branch. Proposition 5.20 gives the same dichotomy for every fixed wall budget.

Unbounded-wall branch. Lemma 5.21 supplies linearly many bounded-wall same-signature atoms, with lower-rank endpoint systems. Bounded carrier corrections belong to the finite palette of Definition 5.22; unbounded corrections localize to a lower-rank wall-free strip.

Translated-arch branch. Lemma 5.23 gives either immediate anchored deletion or one fixed-coordinate endpoint/carrier escape. The latter is lower rank unless the only remaining mass is the activation/debt configuration.

Activation/contact branch. Sections 5.8–5.13 derive the activation mechanism from exact translated-arch endpoint identities, reduce the terminal survivor to a bounded high contact, and obtain the actual contact-centered relative valley of Theorem 5.38. Section 5.14 then extracts literal cycle packages from the observed synchronized trace, without trace-changing flattening. Bounded activation enters a finite projective system. Unbounded activation produces an actual threshold contact, and Lemma 5.44 prevents return to the same-rank superlevel branch. The return therefore ends in finite-projective compactness, support/provenance decrease, or a genuine lower-rank projective/relative escape.

All branches either terminate in Proposition 5.13, strictly lower recurrent guard rank, strictly lower provenance rank, or advance to a later same- (ν, ρ) stage in the finite stage order made explicit in Table 1. A responsible escape is therefore not a same-state recursive call: at its first occurrence it is reclassified as one of these successor types. Lemma 5.44 rules out the only possible return from the contact stage to the earlier activation stage at unchanged (ν, ρ) . Hence the rank- ρ bounded-context factor is structurally reduced to terminal compactness or a strictly smaller proof state. The computability of the resulting numerical bound is established separately in Proposition 5.47. $\square$

5.17 Quantitative effectivity and the proof-state order

The preceding theorem is structural. We now explain why its bound is a total computable integer rather than only an abstract finiteness statement.

Every tracked integer observable changes by at most a computable constant on the finite primitive macro alphabet. Therefore, for any cutoff M , the *first* exit from $[-M, M]$ occurs in a finite band

$$M < |Z| \leq M + D$$

for a computable drift bound D . Before this first exit the factor lies in a finite normalized context box, so the relevant lower-rank WCI bound applies. Wall-density constants are explicit, exact wall and carrier colors are finite, and every Ramsey or pigeonhole refinement is invoked only after choosing a finite required chain length.

A potentially dangerous infinitary Dickson extraction is avoided. Along a strict endpoint chain, after choosing a fixed reference coordinate, one obtains

$$(a_0 - a_n) + (\theta_0(q) - \theta_n(q)) \geq n. \tag{31}$$

Thus either common anchored scalar debt is large or projective spread is large. Projective spread above the explicit synchronized-product simple-path bound exposes a bounded nonzero differential core; below that bound normalized profiles lie in a finite box.

For complete algorithmic bookkeeping define the lexicographic measure

$$\mathcal{M} = (\nu, \rho, s),$$

where ν is the provenance rank from Definition 5.9, ρ is recurrent guard rank, and s belongs to the following finite stage set, ordered from top to bottom. Support changes that preserve the represented provenance atoms are finite-mode refinements; a recursive “support/provenance drop” means a strict decrease of ν as in Lemma 5.10.

s	Stage	Allowed same- (ν, ρ) successor
7	bounded-context entry/refinement	wall decomposition or terminal finite projective
6	directional wall decomposition	fixed- K terminal or unbounded-wall skeleton
5	dense same-signature atoms	translated high-wall carrier stage
4	translated arches / vector shell defect	anchored deletion, lower rank, or activation cells
3	activation cells / bounded contact / actual trace	relative valley or lower-rank terminal
2	cycle package / activation gain	critical deletion, finite-projective phase, or threshold contact
1	contact return	finite-projective terminal or strict lower-rank/support escape
0	terminal compactness	no recursive same-rank call

Table 1: Finite same-rank stage component of the proof-state measure.

A genuine provenance-rank drop strictly decreases ν by Lemma 5.10; wall-free responsible escapes strictly decrease ρ ; every other nonterminal transition in Table 1 decreases s . Lemma 5.44 is precisely what prevents a contact return from jumping back from stage 1 to stage 4 at the same (ν, ρ) .

Proposition 5.47 (Effective proof recursion). *For a concrete exact first-return entry, all thresholds and terminal bounds can be computed by well-founded recursion on $\mathcal{M}$. In particular, the bound $B_{\text{FR}}(A, N, C_0, H_0)$ is total computable.*

Proof. The terminal modules are effective: rank at most one uses Proposition 5.17; finite-projective branches use Proposition 5.13; fixed owner-resolved bounded-expression route schemas use Proposition 5.11; fixed wall and carrier alphabets are explicit finite sets. At a nonterminal proof state, compute the finitely many smaller-measure successor bounds first, choose the finite first-exit/Ramsey/wall thresholds required to force one of those successors, and take the finite maximum/sum dictated by the decomposition. Lexicographic well-foundedness guarantees termination. $\square$

Theorem 5.2 follows from Theorem 5.46 and Proposition 5.47.

6 Effective next-zero successors

Fix a concrete zero state

$$Z = (p, H),$$

meaning deterministic configuration $(p, 0)$ and exact net envelope H . The first-return theorem is node dependent: the bound used at Z is computed from this exact H , not from a global uniform envelope bound.

Definition 6.1 (Next-zero factor). A nonempty word u is a next-zero factor from Z if the unique A -run starts at $(p, 0)$, ends at some $(p', 0)$, and has no zero cut at a strict internal word boundary.

Thus a next-zero factor is either a one-letter zero-to-zero transition or a nontrivial positive first return. Put

$$B_0(Z) = \max\{1, B_{\text{FR}}(A, N, Z) + 2\},$$

where the harmless $+2$ absorbs whether the boundary exit/return letters are counted inside the local first-return convention.

For a fixed finite word $u = a_1 \cdots a_k$, the exact transformer $T_u(H)$ is computable without any reachability oracle. For each source coordinate q and each length- k N -path π labelled by u , let $\text{eff}(\pi)$ be its total counter effect and let

$$\text{req}(\pi) = \max\{0, -\min \text{ prefix effect of } \pi\}$$

be the least initial credit required to execute it. Then

$$(T_u H)(r) = \max_{\substack{q, \pi: q \xrightarrow{u} r \\ H(q) \geq \text{req}(\pi)}} (H(q) + \text{eff}(\pi)).$$

The maximum is over a finite set of paths.

Definition 6.2 (Raw zero-successor set). $\text{RawSucc}_0(Z)$ is the set of records

$$e = (u, |u|, p_u, H_u)$$

obtained by enumerating all words $u \in \Sigma^{1 \dots B_0(Z)}$, retaining exactly the next-zero factors, and putting $H_u = T_u(H)$ and $(p_u, 0)$ for the exact deterministic exit.

Theorem 6.3 (Effective zero-successor enumeration). $\text{RawSucc}_0(Z)$ is finite, effectively computable, and literally complete for consecutive zero cuts of every shortest counterexample passing through Z .

Proof. The enumeration is finite. Exact deterministic simulation decides the next-zero property and the formula above computes H_u . If u is the actual factor to the next zero cut of a shortest counterexample, then either $|u| = 1$ or Theorem 5.2 bounds its first-return length by B_{FR} ; hence u occurs in the enumeration. $\square$

There is also a safe finite pruning order, useful for implementations though not needed in the global proof.

Definition 6.4 (Cost-envelope order). For two raw edges with the same target zero control,

$$e_i = (u_i, \ell_i, p', H_i),$$

write $e_1 \preceq_0 e_2$ when $\ell_1 \leq \ell_2$ and $H_1 \leq H_2$ componentwise.

Lemma 6.5 (Zero-edge domination). If $e_1 \preceq_0 e_2$ and $\ell_1 < \ell_2$, then e_2 cannot be used between consecutive zero cuts of a shortest counterexample. If $\ell_1 = \ell_2$, replacing e_2 by e_1 preserves shortestness and rejection.

Proof. Both factors start from the same zero configuration and end in the same deterministic zero control. The smaller envelope preserves rejection by Lemma 5.5. A strict cost decrease contradicts shortestness; equal cost gives another shortest witness. $\square$

The finite set of $\preceq_0$ -minimal edges may therefore be used as a canonical successor antichain when only existence of a shortest witness matters. For the global tree below we keep the raw set, so the actual shortest-witness zero prefix embeds literally rather than only up to replacement.

7 The finite zero-state tree

The local bound of Section 6 is converted into a global bound by a second well-quasi-order argument.

Definition 7.1 (Zero-state order). For zero states (p, H) and (p', H') put

$$(p, H) \leq_Z (p', H') \iff p = p' \text{ and } H \leq H'.$$

Encode $\perp$ by 0 and a finite value h by $h + 1$. For each fixed control p , this is isomorphic to componentwise order on $\mathbb{N}^{|Q_N|}$.

Hence the finite disjoint union of all zero-control components is a WQO by Dickson's lemma.

Construct a rooted edge-labelled tree T_0 as follows. The root is the initial exact zero state. At a node Z , compute $\text{RawSucc}_0(Z)$. For a candidate child $Z' = (p', H')$, inspect only the ancestors on the current root-to-node path. If some ancestor (p', H_a) satisfies $H_a \leq H'$, prune Z' ; otherwise insert it and retain the actual edge word and its cost.

Lemma 7.2 (No infinite branch). *The pruned zero tree has no infinite branch.*

Proof. Along a branch no later zero state is above an earlier zero state with the same deterministic control. If the branch were infinite, some control would occur infinitely often. Dickson's lemma applied to the corresponding infinite sequence of encoded envelopes would produce an increasing pair, contradicting the pruning rule. $\square$

Theorem 7.3 (Finite zero tree). *The pruned tree T_0 is finite and effectively constructible. The complete zero-cut sequence of every shortest counterexample embeds literally as one root-to-node path of T_0 .*

Proof. Theorem 6.3 gives finite branching at every generated node. Lemma 7.2 rules out infinite branches. König's lemma therefore implies finite depth; finite branching at each finite depth gives a finite tree. Exhaustive BFS or DFS terminates because it is enumerating this finite object.

For literal completeness, let $Z_0, \dots, Z_m$ be the actual zero states of a shortest counterexample. Theorem 6.3 inserts the actual factor $Z_i \rightarrow Z_{i+1}$ whenever Z_i is reached. Lemma 4.1 says that no Z_j can be pruned against an earlier Z_i with the same control, because such a pair would itself shorten the witness. Induction on i embeds the whole zero prefix. $\square$

Two computable quantities are extracted from the explicit finite tree:

$$D_0 = \max\{\text{root-to-node edge depth}\},$$

and, retaining literal edge costs,

$$L_0 = \max\{\text{root-to-node total input length}\}.$$

Thus every shortest counterexample has at most $D_0 + 1$ zero cuts, and its prefix through the last zero cut has length at most L_0 .

Remark 7.4 (Why König is effective here). König's lemma is used only to prove that the explicitly generated finitely branching tree is finite. The algorithm itself is ordinary exhaustive generation; it does not require a closed-form Dickson length function. Once mathematical finiteness is established, the generation necessarily reaches an empty frontier.

8 The terminal positive tail

A shortest counterexample may end strictly after its last zero cut. This case has no final zero return, so it is not directly covered by Theorem 5.2. A fresh-symbol construction turns it into one.

Fix a zero node $Z = (p, H)$ of T_0 .

Definition 8.1 (Positive terminal tail). A word u is a positive terminal tail from Z if the A -run starts in $(p, 0)$, is positive at every noninitial cut, and ends in an accepting control at height $h \geq 1$. It is rejecting if N has no accepting continuation over u from exact input envelope H .

Introduce three fresh letters $\#, d, \$$. Form A_Z^{tail} with a start-zero copy of p and positive-phase copies q^+ of the original controls. The start copy permits exactly the original zero rules that leave zero; the positive copies permit exactly the original positive rules. From each accepting f^+ add

$$f^+ \xrightarrow{\#, 0} r,$$

where r is a drain control. At r add the positive rule $r \xrightarrow{d, -1} r$ and the zero rule $r \xrightarrow{\$, 0} \text{acc}$. The only accepting control is acc .

Extend N to N^{tail} by adding zero-effect self-loops on $\#, d, \$$ at every state, leaving the final states unchanged.

Lemma 8.2 (Identity of the fresh suffix). *For every original-alphabet word u and every $k \geq 0$,*

$$T_{u\#d^k\$}^{N^{\text{tail}}}(H) = T_u^N(H).$$

In particular the extended net accepts $u\#d^k\$$ from H iff the original net accepts u from H .

Proof. Each fresh letter has exactly one zero-effect self-loop at every net control and no other fresh-letter transition, so its exact envelope transformer is the identity. $\square$

Lemma 8.3 (Exact tail/corridor correspondence). *A rejecting positive terminal tail u from Z ending at deterministic height h exists iff*

$$u\#d^h\$ \in L(A_Z^{\text{tail}}) \setminus L_H(N^{\text{tail}}),$$

where the net side starts from envelope H . Every accepted word of the extended deterministic machine has this unique form.

Proof. The extended control graph forces the original positive phase, then the marker from an accepting original control, then exactly h decrements to reach zero, then the zero-only final marker. Lemma 8.2 gives the net equivalence. $\square$

Theorem 8.4 (Terminal-tail reduction). *For every zero node $Z = (p, H)$ there is a total computable integer $B_{\text{tail}}(Z)$ such that, if any rejecting positive terminal tail exists from Z , then one exists with length at most $B_{\text{tail}}(Z)$.*

Proof. If a rejecting tail exists, choose a shortest extended counterexample $u\#d^h\$$ given by Lemma 8.3. The factor $u\#d^h$ is a strictly positive first return from zero to zero: the original part stays positive, the marker has zero effect, and only the last drain letter reaches zero. Therefore Theorem 5.2 applied to the extended instance gives

$$|u\#d^h| \leq B_{\text{FR}}(A_Z^{\text{tail}}, N^{\text{tail}}, (s_p, 0), H).$$

Take this first-return bound as $B_{\text{tail}}(Z)$; then $|u| \leq B_{\text{tail}}(Z)$. $\square$

The theorem is existential: it need not bound the literal terminal suffix of every shortest original witness. For global decidability it is enough that any obstruction at the same exact last-zero state has a bounded replacement tail.

Because T_0 is finite, compute

$$B_{\text{tail}}^{\max} = \max_{Z \in T_0} B_{\text{tail}}(Z).$$

9 Global small witnesses and decidability

Define

$$B_{\text{global}} = L_0 + B_{\text{tail}}^{\max}.$$

Theorem 9.1 (Global small witness, detailed form). *If $L(A) \not\subseteq L(N)$, then there is a word $w \in L(A) \setminus L(N)$ with*

$$|w| \leq B_{\text{global}}.$$

Proof. Choose a shortest counterexample. If it ends at a zero cut, its entire word is a root-to-node path in T_0 , hence has length at most L_0 . Otherwise write it xu , where x ends at the last zero state $Z = (p, H)$ and u is a rejecting positive terminal tail. The literal zero prefix x embeds in T_0 , so $|x| \leq L_0$. Theorem 8.4 supplies another rejecting positive tail v from the same exact deterministic configuration $(p, 0)$ and the same exact net envelope H , with $|v| \leq B_{\text{tail}}(Z)$. Consequently xv is still accepted by A and rejected by N , and

$$|xv| \leq L_0 + B_{\text{tail}}^{\max}.$$

□

Corollary 9.2 (Hard inclusion). *It is decidable whether $L(A) \subseteq L(N)$.*

Proof. Compute the finite integer B_{global} and enumerate $\Sigma^{\leq B_{\text{global}}}$. Deterministic membership in A is direct. Membership in N is a finite dynamic program over the word. If one enumerated word lies in $L(A) \setminus L(N)$, report noninclusion; if none does, Theorem 9.1 proves inclusion. □

10 The reverse inclusion

The hard part of the paper is $L(A) \subseteq L(N)$. Equality also requires the opposite orientation; we spell out the reduction because the deterministic machine has zero tests.

Proposition 10.1 (Deterministic-right inclusion). *It is decidable whether $L(N) \subseteq L(A)$.*

Proof. 1. Total deterministic complement. The deterministic OCA A may be partial. Add one dead control $\dagger$ and, for every missing letter transition in either the zero or positive regime, add a zero-effect transition to $\dagger$; add zero-effect self-loops on all letters in both regimes at $\dagger$. The resulting deterministic OCA is total on every word. Complement the set of final controls to obtain $\bar{A}$, so

$$L(\bar{A}) = \Sigma^* \setminus L(A).$$

2. Synchronous product. Now

$$L(N) \not\subseteq L(A) \iff L(N) \cap L(\bar{A}) \neq \emptyset.$$

Synchronize one transition of N with the unique $\bar{A}$ transition on each letter. The product has two nonnegative counters: the N counter is never zero-tested and the A counter is the only tested counter.

3. Compile the positive regime. A positive-regime OCA rule must be disabled at A -counter value zero. Simulate it by first decrementing the A counter by one, which is possible exactly when it is positive, while performing the synchronized N update. Then use a bounded finite-control chain to add back the amount needed to realize the original update $-1, 0$, or $+1$ (equivalently use the corresponding fixed VASS vector if arbitrary finite vectors are allowed). Thus positivity is enforced without any zero test.

4. Serialize all zero-regime rules through one zero-test gate. The product may contain many zero-regime rules, whereas the standard VAS_0 presentation uses one distinguished zero-test transition. This difference is harmless and we spell out the reduction. For every product zero-rule e , introduce one fresh ordinary tag counter t_e . From the source control of e , an ordinary transition moves to a common pre-gate control and places one token in t_e , without changing the tested A counter. A single distinguished transition

$$g_0 : q_{?0} \longrightarrow q_{=0}$$

is enabled exactly when the A counter is zero. After g_0 , the unique post-gate transition corresponding to e consumes the token in t_e , performs the synchronized N update and the legal zero-regime A update (necessarily 0 or $+1$ in unit-step form), and enters the target product control. All other post-gate branches are disabled because their tag counter is zero. If the source A counter was positive, the run can enter the pre-gate gadget but cannot pass g_0 , so it produces no spurious reachable target. Hence every zero-regime product edge is simulated exactly while all tests share one single distinguished zero-test transition.

The resulting counter system is therefore effectively a VASS/VAS with one zero-test in the sense of Reinhardt and Bonnet, for which exact configuration reachability is decidable [10, 11].

5. Final counters need not be zero. The product accepts upon reaching a control pair in $F_N \times F_{\bar{A}}$ with arbitrary nonnegative counter values. From every such product control enter a drain phase with ordinary decrement loops for each counter. Then some accepting product control is reachable with arbitrary counters iff one fixed drain target with both counters equal to zero is exactly reachable. No additional zero test is required because exact reachability asks for the target vector itself.

Therefore noninclusion is decidable, and so is $L(N) \subseteq L(A)$. □

Combining Proposition 10.1 and Corollary 9.2 proves Theorem 1.1.

11 The epsilon-move extension: models and deterministic reset normalization

11.1 Deterministic epsilon one-counter automata

A deterministic epsilon one-counter automaton (epsilon-OCA) has a finite control set Q_A , an initial control q_A^0 , a set F_A of accepting controls, a nonnegative counter initialized to 0, a visible alphabet Σ , and transitions labelled by $\Sigma \cup \{\varepsilon\}$. Transitions may distinguish counter zero from counter positive. We use the standard unit-update OCA model, with counter effects in $\{-1, 0, +1\}$. This unit-update convention is part of the theorem statement; no unproved expansion of larger deterministic updates is used below.

The determinism convention is the standard one: in a fixed control/sign regime, if an epsilon transition is enabled then no visible transition competes with it. A visible word is accepted when the unique complete deterministic computation is defined, includes all forced silent behavior, and ends in a control of F_A ; the final counter may be any nonnegative integer. If the computation becomes undefined, the word is rejected.

We use the standard stable/reset normalization of deterministic epsilon-OCA from the deterministic one-counter equivalence literature [5]. In this presentation, stable controls consume

visible letters and reset controls perform forced silent behavior. A reset control s carries a finite period per_s and a map

$$\text{goto}_s : \{0, \dots, \text{per}_s - 1\} \rightarrow Q_A^{\text{st}},$$

with semantics

$$(s, n) \xRightarrow{\epsilon^*} (\text{goto}_s(n \bmod \text{per}_s), 0).$$

Initial and accepting configurations are arranged stable. We take this normalization as an external standard input and perform the remainder of the hard-direction proof in the stable/reset presentation.

Remark 11.1. The deterministic side of Theorem 1.3 is not allowed arbitrary nondeterministic silent behavior. The theorem covers the standard deterministic-epsilon convention described above.

11.2 Nondeterministic epsilon one-counter nets

A nondeterministic epsilon one-counter net (epsilon-OCN) has a finite control set Q_N , an initial control q_N^0 , a set F_N of accepting controls, one nonnegative counter initialized to 0, visible and epsilon transitions, and no zero test. Primitive transitions have unit effects $d \in \{-1, 0, +1\}$ and are enabled exactly when the resulting counter remains nonnegative. Weak macro effects introduced below are arbitrary integers because they sum finitely many primitive effects. A word is accepted when some weak run ends in a control of F_N , with arbitrary nonnegative final counter value.

For a visible word w , a weak run is any run whose visible projection is exactly w ; arbitrary epsilon moves may occur before, between, and after visible letters. The empty visible word is included.

11.3 Path requirement and effect

For a finite OCN path π with counter effects $d_1, \dots, d_m$, define

$$\text{eff}(\pi) = \sum_{i=1}^m d_i$$

and

$$\text{req}(\pi) = \max \left(0, - \min_{0 \leq j \leq m} \sum_{i=1}^j d_i \right).$$

Lemma 11.2. *A path π is feasible from counter c if and only if $c \geq \text{req}(\pi)$. In that case its exit counter is $c + \text{eff}(\pi)$.*

Proof. Let $s_j = \sum_{i=1}^j d_i$, with $s_0 = 0$. Starting from counter c , the counter after the first j transitions is $c + s_j$. Thus the whole path is feasible exactly when $c + s_j \geq 0$ for every j , equivalently when $c \geq -\min_j s_j$ and $c \geq 0$. This is precisely $c \geq \text{req}(\pi)$. The final counter is $c + s_m = c + \text{eff}(\pi)$. $\square$

This elementary formula is the basis of the weak-envelope construction.

11.4 Stable cuts and visible length

A stable cut is taken after one visible input letter has been consumed and every forced deterministic reset closure triggered by that letter has completed. A stable zero configuration is a stable deterministic configuration $(p, 0)$.

All length bounds in the paper count visible input letters only. Silent microsteps have zero word cost.

12 Weak epsilon-OCN envelope calculus

For controls $p, q \in Q_N$ and $a \in \Sigma$, write

$$(p, c) \Rightarrow_a (q, d)$$

when there is an OCN path of shape

$$\varepsilon^* a \varepsilon^*$$

from (p, c) to (q, d) . Define

$$U_{p,a,q}(c) = \sup\{d : (p, c) \Rightarrow_a (q, d)\},$$

with value $\perp$ when no such weak path exists and ω when reachable exit counters are unbounded.

12.1 Cycle deletion and pumping

Lemma 12.1 (Nonpositive epsilon-cycle deletion). *If a weak- a path contains an epsilon cycle C with $\text{eff}(C) \leq 0$, deleting that literal occurrence of C does not increase the path's entry requirement and does not decrease its exit counter.*

Proof. Write the path as $\alpha C \beta$ and let $e = \text{eff}(C) \leq 0$. Prefix sums before C are unchanged after deletion. Prefix sums after the deleted block increase by $-e \geq 0$, while the internal prefixes of C disappear. Hence the minimum prefix sum cannot decrease, so the entry requirement cannot increase. The total effect increases by $-e$, and therefore the exit counter does not decrease. $\square$

Lemma 12.2 (Positive epsilon-cycle pumping). *If a feasible weak- a path contains an epsilon cycle C with $\text{eff}(C) > 0$ in its epsilon prefix or suffix, then above an effectively computable entry threshold the corresponding target upper envelope is unbounded.*

Proof. Write one weak path containing the positive epsilon cycle as $\rho C \tau$, where C is silent and $g = \text{eff}(C) > 0$. Put

$$r_C = \text{req}(\rho C).$$

For every source credit $c \geq r_C$, the prefix ρ and one traversal of C are feasible. After that traversal the counter at the cycle base has increased by g , so the same cycle remains feasible; after k repetitions it has gained kg . The fixed continuation τ has a finite entry requirement, hence sufficiently many repetitions make τ feasible and the target counter can be made arbitrarily large. Thus r_C is the exact entry threshold for this fixed pump witness.

Conversely, if pumping is possible from a source credit c , some feasible weak run contains a positive epsilon closed walk. Decomposing that walk into simple cycles yields a simple positive epsilon cycle. Choose the first such cycle on the run and delete all preceding nonpositive silent cycles; this produces one of the finitely enumerable pump witnesses with $r_C \leq c$. Taking the minimum r_C over all such witnesses therefore gives the exact onset of the pumping branch. $\square$

Theorem 12.3 (Finite weak-envelope normal form). *For every p, a, q one can effectively compute:*

- an optional pumping threshold r_∞ , and
 - a finite set $S_{p,a,q} \subseteq \mathbb{N} \times \mathbb{Z}$ of threshold/effect pairs,
- such that

$$U_{p,a,q}(c) = \omega$$

whenever the pump threshold applies, and otherwise

$$U_{p,a,q}(c) = \max\{c + e : (r, e) \in S_{p,a,q}, r \leq c\},$$

with $\perp$ when the set of enabled clauses is empty.

Proof. Positive epsilon cycles give the pumping branch by Theorem 12.2. Below all pump thresholds, take any feasible weak a -path and repeatedly delete every nonpositive epsilon cycle using Theorem 12.1. If an epsilon prefix or suffix still contained a closed walk, it would contain a simple cycle; because positive cycles are excluded below the pump threshold, such a cycle would be nonpositive and could be deleted. Thus both silent parts become simple. There are only finitely many resulting simple $\varepsilon^* a \varepsilon^*$ skeletons. By Theorem 11.2, each skeleton contributes one pair $(\text{req}(\pi), \text{eff}(\pi))$, and every deleted path is dominated by its reduced skeleton. Taking the maximum over enabled skeletons gives the displayed finite branch. Enumerating simple connectors and simple positive epsilon cycles makes both the finite clause set and the pumping thresholds effective; Appendix D gives the explicit construction. $\square$

For all later proof bookkeeping we use a *skeleton-refined palette*: if several simple weak paths induce the same threshold/effect pair, they are retained as distinct macro types whenever their concrete skeleton identity is needed. Likewise a pump type retains one concrete lasso certificate (prefix, positive silent cycle, and suffix) witnessing its threshold. This finite refinement does not change the envelope transformer.

12.2 Forward envelopes and backward requirements

Let

$$D = \{\perp\} \cup \mathbb{N} \cup \{\omega\},$$

ordered by

$$\perp < 0 < 1 < 2 < \dots < \omega.$$

A forward envelope is a vector $H \in D^{Q_N}$. The value $H(q) = \omega$ means that arbitrarily large finite counters are reachable at q ; it does not mean that every sufficiently large integer is reachable.

For $a \in \Sigma$, define

$$T_a^\varepsilon(H)(q) = \max_{p \in Q_N} U_{p,a,q}(H(p)),$$

with the natural interpretation of a source value ω through the finite threshold/pump description.

For a visible suffix v and state q , define

$$R_v(q) = \min\{c \in \mathbb{N} : \text{from } (q, c) \text{ some weak run reading } v \text{ reaches a final control}\},$$

with $R_v(q) = \infty$ when the set is empty.

Definition 12.4 (Acceptance predicate). For a forward envelope H and requirement vector R , write $\text{Acc}(H, R)$ if there exists q such that $R(q) < \infty$ and either $H(q) = \omega$ or $H(q) \in \mathbb{N}$ with $H(q) \geq R(q)$. Write $\text{Rej}(H, R)$ for the negation.

We deliberately do not encode rejection by a global expression $H < R$ once ω and ∞ coexist.

Theorem 12.5 (Exact weak residuation). *For every visible letter a , there is an effective backward transformer S_a^ε such that*

$$\text{Acc}(T_a^\varepsilon(H), R) \iff \text{Acc}(H, S_a^\varepsilon(R)).$$

Equivalently,

$$\text{Rej}(T_a^\varepsilon(H), R) \iff \text{Rej}(H, S_a^\varepsilon(R)).$$

Proof. For each source control p , define $S_a^\varepsilon(R)(p)$ semantically as the least finite counter from which some weak a -run reaches a target q with enough exit credit to satisfy the finite requirement $R(q)$; use ∞ if no such counter exists. The feasible source set is upward closed because $\mathbb{N}$ has no zero tests. Therefore a forward envelope H can realize an R -accepting weak a -continuation

exactly when one of its coordinates meets this least source requirement; an ω coordinate meets every finite requirement because it represents unbounded reachable finite credit. This is precisely the displayed equivalence. Effectivity follows from the finite simple-skeleton and pump description of Theorem 12.3; the concrete formula is recorded below and in Appendix D. $\square$

For a finite simple weak skeleton $\pi : p \Rightarrow_a q$ and finite target requirement $R(q)$, the corresponding source requirement is

$$\max(\text{req}(\pi), R(q) - \text{eff}(\pi), 0).$$

A pump lasso contributes one finite entry threshold independent of the magnitude of the finite target requirement.

12.3 Initial and terminal epsilon closure

Let H_ϵ be the exact epsilon saturation of the initial N configuration. Let

$$R_\epsilon(q)$$

be the least counter from which an epsilon-only run reaches an accepting control, or ∞ if impossible. If $w = a_1 \cdots a_n$, then

$$H_w = T_{a_n}^\epsilon \cdots T_{a_1}^\epsilon(H_\epsilon),$$

and

$$w \in L(N) \iff \text{Acc}(H_w, R_\epsilon).$$

The same formula handles the empty visible word.

Lemma 12.6 (Active-omega exclusion). *If uv is rejected and $R_v(q) < \infty$, then $H_u(q) \neq \omega$.*

Proof. If $H_u(q) = \omega$, arbitrarily large finite counters are reachable at q after u . Since $R_v(q) < \infty$, choose one reachable counter $c \geq R_v(q)$. By definition of $R_v(q)$, the suffix v then has an accepting weak continuation, contradicting rejection of uv . $\square$

13 From unit edges to threshold-affine weak macros

This section records the epsilon-specific transfer lemmas needed to lift the real-time positive first-return compactness theorem.

13.1 Epsilon separator margin

For $c \in \mathbb{N}$, define $H + c$ by leaving $\perp$ and ω unchanged and adding c to finite coordinates. At a rejected cut set

$$m_\epsilon(H, R) = \min\{c \in \mathbb{N} : \text{Acc}(H + c, R)\},$$

with value ∞ if no such c exists.

Lemma 13.1 (Weak shift inequality). *For every visible factor x and $c \geq 0$,*

$$T_x^\epsilon(H + c) \geq T_x^\epsilon(H) + c.$$

Proof. Every concrete weak run feasible from a finite source counter h remains feasible from $h + c$ and exits exactly c higher. Taking coordinatewise suprema preserves the inequality. If a coordinate on the right is ω , then arbitrarily large finite outputs from H remain arbitrarily large after the source shift, so the left coordinate is also ω . $\square$

Lemma 13.2 (Margin monotonicity). *If*

$$H' = T_x^\varepsilon(H), \quad R = S_x^\varepsilon(R'),$$

and the factor lies in a rejected word, then

$$m_\varepsilon(H, R) \leq m_\varepsilon(H', R').$$

Proof. Let $c = m_\varepsilon(H', R') < \infty$. Then $\text{Acc}(H' + c, R')$. By Theorem 13.1, $T_x^\varepsilon(H + c) \geq H' + c$, and acceptance is monotone in the forward envelope. Hence $\text{Acc}(T_x^\varepsilon(H + c), R')$. Applying exact residuation Theorem 12.5 gives $\text{Acc}(H + c, S_x^\varepsilon(R')) = \text{Acc}(H + c, R)$, so $m_\varepsilon(H, R) \leq c$. The case $c = \infty$ is immediate. $\square$

13.2 Threshold contacts and protected propagation

A finite weak macro is a tuple

$$e = (p, a, q, r, \delta, \sigma),$$

where r is its entry threshold, δ its integer effect, and σ identifies one concrete simple weak skeleton $\varepsilon^* a \varepsilon^*$ realizing the clause. Pump clauses are recorded separately.

Lemma 13.3 (Pump-or-threshold contact). *Suppose a state q is critical at one step of a constant finite separator margin $\eta \geq 1$. If a backward-minimizing witness is a pump clause with threshold r_{pump} , then*

$$H(q) = r_{\text{pump}} - \eta < r_{\text{pump}}.$$

Hence the source lies in a computably finite threshold band.

Proof. A pump witness can satisfy any finite target requirement once its source counter reaches its fixed threshold, so its backward candidate is exactly r_{pump} . If it attains the backward minimum at the critical source, then $R(q) = r_{\text{pump}}$. Criticality gives $R(q) - H(q) = \eta$, hence $H(q) = r_{\text{pump}} - \eta < r_{\text{pump}}$. $\square$

Lemma 13.4 (Protected finite-macro propagation). *Suppose a backward-minimizing witness is a finite macro with threshold r and the critical source satisfies $H(q) \geq r$. Then the macro is forward-enabled, forward-maximizing, backward-tight, and transports the critical gap exactly to the next cut.*

Proof. Let the macro go from q to q' with effect δ , and let R' be the next-cut requirement. At the critical source $R(q) = H(q) + \eta > H(q) \geq r$. Since the macro attains the backward minimum, its candidate is $\max(r, R'(q') - \delta) = R(q)$. The maximum therefore cannot be attained by r , and so $R(q) = R'(q') - \delta$. The source threshold is satisfied, hence the macro is forward-enabled and

$$H'(q') \geq H(q) + \delta = R'(q') - \eta.$$

Constant separator margin η at the next cut gives the opposite inequality $H'(q') \leq R'(q') - \eta$. Thus equality holds: the macro is forward-maximizing, the backward equality is tight, and q' is critical with the same gap η . $\square$

Lemma 13.5 (Threshold-contact recurrence is margin-independent). *Work inside one constant-margin phase. If a backward-minimizing finite macro with threshold r is not forward-enabled at a critical source, then its source credit is one of the finitely many values*

$$c = H(q) \in \{0, \dots, r - 1\}.$$

For recurrence/pigeonhole bookkeeping it is sufficient to color such a contact by the refined finite mode, the critical source state, the chosen macro type, the bounded value c , and the finite

support/omega data. The numerical separator margin η need not be included in the contact color. Consequently, if deterministic height and projective spread relative to the pinned value c are bounded, only finitely many exact contact states occur; a repeated exact state admits the usual shortest-witness deletion. All contact multiplicity bounds are therefore independent of the future backward requirement.

Proof. The inequality $H(q) < r$ gives the finite source-credit set. Inside the chosen phase the separator margin is already constant, so coloring by its numerical value contributes nothing to identifying the forward exact state. After fixing the finite mode and omega mask, bounded deterministic height and bounded projective spread around the pinned finite coordinate c bound every remaining finite envelope coordinate. Hence the exact state space is finite. If the same deterministic configuration and exact envelope recur, deleting the intervening visible factor rejoins both machines at the same exact forward state and preserves rejection against the inherited suffix, contradicting shortestness. No numerical bound on η is used. $\square$

Thus every constant-margin factor either exposes a bounded threshold contact or carries a coherent tight critical strand through finite weak macros.

13.3 High-zone pump obstruction

Lemma 13.6 (High-zone pump obstruction). *Inside a bounded projective/relative mode, a pump macro into a backward-live target is either disabled below a computable finite deterministic-height threshold or contradicts rejection. Consequently, above a computable height, all backward-live active coordinates evolve only by finite threshold-affine macros.*

Proof. In a fixed bounded projective/relative mode, every finite present source has form $H(p) = d + \kappa_p$, where d is the deterministic height parameter and κ_p ranges over a finite computable set. Consider a pump macro from p to a target q with threshold r_p . If q is backward-live and the pump is enabled, then the next forward value at q is ω , contrary to Theorem 12.6 for a rejected continuation. Hence every such pump must satisfy $d + \kappa_p < r_p$, which bounds d by $r_p - 1 - \kappa_p$. Taking the maximum over the finite pump palette gives the claimed computable high-zone cutoff. Above it, only finite affine macros can contribute to backward-live targets. $\square$

13.4 First amplification and trace fidelity

Lemma 13.7 (Regime nonexpansiveness and first split). *Fix a visible letter a . Let H, K have the same finite support/omega mask and enable the same finite and pump macro clauses. On every target coordinate that is finite in both outputs,*

$$|T_a^\varepsilon(H)(q) - T_a^\varepsilon(K)(q)| \leq \|H - K\|_\infty$$

where the norm ranges over their common finite source coordinates. Moreover, if two previously identical finite semantic regimes first diverge on the next visible step, then the divergence is caused by a support/omega event or by crossing one fixed finite or pump threshold; if the pre-split finite sup-distance is at most L , the responsible source values lie in a computably bounded L -neighborhood of that threshold.

Proof. For a finite target in a fixed regime, the weak transformer is the maximum of the same finite family of affine candidates $H(p) + \delta_e$; taking a maximum of functions with slope one in one source coordinate is 1-Lipschitz in the sup norm. Enabled pump clauses produce ω on both sides and therefore do not enter the finite-coordinate estimate. If the next regimes differ for the first time, the two inputs had the same support/omega mask and the same clause status immediately before that step. Hence the first disagreement is either a support/omega event or the enabling status of some clause with fixed threshold r . In the latter case one source value is

below r and the other is at least r ; a pre-split distance at most L places both values in the finite band $[r - L, r + L]$ after intersecting with $\mathbb{N}$. $\square$

Lemma 13.8 (Exact finite-composition closure). *Fix a refined route $e_1 \cdots e_k$ of weak macro choices with matching controls. If every e_j is finite, write r_j, δ_j for its threshold and effect and $E_{j-1} = \sum_{i < j} \delta_i$. Then the composition is exactly one threshold-affine route*

$$c \mapsto c + \sum_{j=1}^k \delta_j \quad \text{for } c \geq r_{\text{comp}} := \max\left(0, \max_{1 \leq j \leq k} (r_j - E_{j-1})\right).$$

If the first pump on the route occurs at position j , then the whole route is a pump route with exact source threshold

$$r_{\text{comp}} = \max\left(0, \max_{i < j} (r_i - E_{i-1}), r_j^{\text{pump}} - E_{j-1}\right).$$

The construction is effective. A finite composition retains the concatenated concrete weak skeleton; a pump composition retains the finite prefix/suffix skeletons together with one concrete positive-cycle lasso certificate.

Proof. Before the first pump, the entry credit at macro e_j is $c + E_{j-1}$, so simultaneous feasibility of all finite prefixes is exactly the displayed maximum of shifted thresholds. With no pump, the endpoint formula follows by summing effects. If the first pump is enabled, it produces arbitrarily large reachable finite credit at its target. Every fixed remaining suffix has only finitely many finite thresholds (or another pump), so sufficiently large pumped credit executes that suffix and leaves unbounded final credit. Conversely, before the first pump no smaller source credit can violate any of the listed prefix/first-pump requirements. The skeleton and lasso certificates are obtained by concatenation. $\square$

Lemma 13.9 (Residue-refined powered-schema realization). *Fix an owner-resolved bounded-expression schema over the skeleton-refined finite weak-macro palette. After an effective finite refinement of every powered exponent into residue classes, its threshold feasibility, owner comparisons, support/mode constraints, and deterministic positivity are Presburger conditions on the exponent variables; on each residue branch the endpoint finite owner vector is affine in the quotient exponent variables. The refinement and all formulas are independent of the future backward requirement.*

Proof. First apply Theorem 13.8 to each fixed connector and powered block. On a fixed owner-resolved block β , the finite owner vector has an affine update $x \mapsto Px + c$ for one effectively known 0–1 functional matrix P . Since the powers of a finite matrix chosen from this finite transformation monoid are eventually periodic, compute μ, λ with $P^{\mu+\lambda} = P^\mu$. For $n = \mu + s + \lambda k$ with $0 \leq s < \lambda$,

$$P^n x + \sum_{j < n} P^j c$$

is a fixed residue-dependent affine function of x and k : the tail of the matrix-power sequence is periodic and each full period contributes one fixed integer vector. The finitely many cases $n < \mu$ are handled separately.

Every macro threshold and every owner comparison is an affine inequality in the current finite owner vector. After splitting the repetition index by the same finite residues, its validity at an intermediate repetition is a Presburger formula in the quotient exponents and the repetition index. Requiring validity for all repetitions is therefore Presburger by effective closure under first-order quantification. Repeating this construction for the finitely many powered blocks and taking the finite product of residue branches proves the claim. No backward requirement appears in the construction. $\square$

For trace arguments, macro types retain the concrete simple skeleton identifier σ . Concatenating selected actual macro occurrences thus gives one actual N micro-run over the visible factor. In particular, an actual synchronized macro cycle consumes a nonempty contiguous visible word. No generic alternative VASS path is used to infer literal structure of the observed word.

14 Guarded-macro robustness of positive first-return compactness

The real-time compactness theorem has already been proved in Theorem 5.2. We now isolate the exact features of that proof that are used by the epsilon compilation. This turns the epsilon extension into a self-contained internal robustness theorem.

Definition 14.1 (Admissible guarded macro system). Fix the two machines and one concrete exact zero entry. An *admissible guarded macro system* is a finite, effectively computed family of visible macro types (for all visible letters, not for one preselected corridor) such that:

- (G1) every finite macro has a source threshold r , an integer endpoint effect δ , and a concrete weak skeleton, and acts as $c \mapsto c + \delta$ whenever $c \geq r$;
- (G2) every pump macro has one finite enabling threshold and one concrete lasso certificate and, when enabled, produces unbounded exit credit, represented by ω ;
- (G3) the induced forward and backward transformers are exact and monotone and satisfy exact residuation; moreover, on a constant-margin rejected phase, every enabled finite backward-minimizing critical macro is simultaneously forward-maximizing and transports the same tight separator equality to the next cut;
- (G4) common-credit shifts are monotone: for every visible factor x and $c \geq 0$, $T_x(H+c) \geq T_x(H)+c$ on finite coordinates, with $\perp$ and ω fixed under the shift;
- (G5) on each fixed finite support/omega/threshold regime the transformer is nonexpansive in finite coordinates, and a first regime split is localized to a support/omega event or to a bounded neighborhood of one of finitely many thresholds; threshold-contact recurrence uses only the bounded source credit and finite mode data, not the numerical separator margin;
- (G6) every tracked integer observable used by the compactness proof has a computable one-macro drift bound; moreover every fixed bounded-expression macro schema $\alpha_0 \beta_1^{n_1} \alpha_1 \cdots \beta_m^{n_m} \alpha_m$ has effectively Presburger threshold/guard/owner feasibility, and after an effective finite exponent-residue refinement its endpoint data are affine in the quotient exponent variables;
- (G7) support, owner/provenance, guard, threshold, and omega data range over finite effectively enumerable mode alphabets, while bounded-height exact-state boxes are well-quasi-ordered after fixing the finite omega mask;
- (G8) every macro used in a literal deletion argument retains a concrete skeleton from the observed weak run, so a selected synchronized macro cycle represents a nonempty contiguous visible factor of the observed word;
- (G9) the palette is effectively closed under every fixed finite composition used as a proof macro: after finitely refining clause and pump choices, each composed route is again either threshold-affine with one computable source threshold and total integer effect, or a pump route; finite routes retain the concatenated concrete skeleton, while pump routes retain a concrete lasso certificate and finite prefix/suffix skeletons.

All palettes, mode alphabets, drift bounds, and thresholds in (G1)–(G9) are computed from the machines and the concrete exact zero entry and are independent of the future suffix and its backward requirement.

Theorem 14.2 (Weak-palette realization). *For the skeleton-refined weak semantics of the epsilon-OCN, the macro palette computed from the machines satisfies (G1)–(G9) of Theorem 14.1. The*

construction is uniform over all ordinary positive first-return corridors from the fixed exact zero entry and is independent of the future suffix, exit requirement, and numerical separator margin.

Proof. The skeleton-refined finite weak-envelope normal form Theorem 12.3 gives (G1) and (G2), including concrete finite skeletons and pump lasso certificates. The exact forward transformer is a finite maximum of those clauses, and Theorem 12.5 gives the exact monotone backward transformer. On a constant-margin rejected phase, Theorem 13.4 shows that every forward-enabled finite backward minimizer is also forward-maximizing and transports the same tight gap; this is precisely (G3). The shift property (G4) is Theorem 13.1.

For (G5), fixed enabled-clause/support/omega regimes are nonexpansive by Theorem 13.7; the first loss of that regime is a support/omega event or a crossing of one of finitely many finite/pump thresholds. Theorem 13.5 shows that recurrent threshold contacts are colored by bounded source credit and finite mode data only—the numerical separator margin is not a color and therefore cannot enter the bound. Pump minimizers are threshold contacts by Theorem 13.3, while Theorem 13.6 removes enabled pumps from backward-live protected high zones.

The finite macro palette gives a computable drift bound for every fixed integer observable. The powered part of (G6) is exactly Theorem 13.9: eventual periodicity of the finite owner-map monoid yields a finite exponent-residue refinement on which endpoint data are affine and all threshold/owner/positivity constraints are Presburger. For (G7), support masks, threshold regimes, owner/provenance frames, guard data, and omega masks range over finite effectively enumerable sets. After fixing the bottom/omega masks, the remaining finite coordinates lie in a finite power of $\mathbb{N}$ and are well-quasi-ordered by Dickson; bounded context boxes are effectively finite when the relevant integer ranges are bounded.

For (G8), every finite target value is a maximum over a finite family of finite skeleton clauses and is therefore attained. Choosing the canonical maximizing predecessor and one attaining skeleton at each observed macro occurrence makes consecutive owner edges concatenate to an actual weak run; no omega coordinate is used on a backward-live rejected strand by Theorem 12.6. Finally, Theorem 13.8 proves (G9), with exact shifted thresholds for finite routes and for the first pump on a pump route. Every construction in this paragraph uses only the machines, the fixed exact entry for its provenance naming, and finite syntactic refinements; no backward requirement or future suffix is an input. $\square$

Theorem 14.3 (Guarded-macro transfer theorem). *Fix a concrete exact zero entry for the deterministic machine and an admissible guarded macro system in the sense of Theorem 14.1. Then the suffix-parametric real-time positive first-return compactness proof of Section 5 remains effective after replacing the unit net edges by these guarded macros. More precisely, one can compute from the machines and the exact entry a single integer B_{GM} such that, for every deterministic zero exit configuration C_1 and every exact backward requirement R at that exit, every strictly positive first-return corridor that is R -minimal in the sense of Theorem 5.1 has visible length at most B_{GM} . In particular, B_{GM} is independent of R and hence of the future suffix.*

Proof. The real-time proof uses only the interfaces listed in (G1)–(G9); the quantitative replacements are recorded in Appendix E. Briefly, (G1), (G6), (G7), and (G9) provide the finite owner/provenance data, additive guarded effects, composition closure, Presburger powered-route checks, and bounded drift; (G4) supplies the common-credit shift inequalities; (G3) and (G5) supply exact critical propagation together with bounded threshold/contact localization; and (G8) preserves literal observed-trace deletion. Hence the wall, carrier, translated-arch, activation, contact, bounded-context, and WQO modules of Sections 5.2–5.16 carry over after finite rescaling.

The terminal modules remain in the same decidability classes: a finite route with threshold r and effect δ is compiled by subtracting r and then adding $r + \delta$, so threshold guards add no zero test to the finite-projective 2-VASS module or to the rank-one one-zero-test module. Pumps

create no new recursive branch: a critical pump is a threshold contact, a protected high-zone pump into a backward-live target contradicts rejection, and an ω coordinate in a rejected factor is backward-dead. Therefore the same lexicographic proof measure (ν, ρ, s) is well founded and the effective recursion of Theorem 5.47 computes one bound from the machines and exact entry alone. Since the real-time theorem is already suffix-parametric, the backward requirement R is never a recursive bound parameter. This yields the stated uniform B_{GM} . $\square$

Theorem 14.4 (Robust positive first-return compactness). *Fix a concrete stable zero entry*

$$Z = (p, H)$$

of the deterministic stable/reset automaton together with the exact ε -saturated net envelope H . There is a total computable integer

$$B_{\text{FR}}^\varepsilon(Z)$$

such that, for every stable zero exit configuration C_1 and every exact backward requirement vector R , every ordinary positive first-return factor P from Z to C_1 that is visible-length-minimal among all words from the same exact entry to C_1 whose exact exit envelope is rejected against R has visible length at most $B_{\text{FR}}^\varepsilon(Z)$. In particular, the same bound holds for every ordinary positive first return occurring inside a globally shortest rejected continuation from Z .

Proof. By Theorem 14.2, the skeleton-refined weak palette is an admissible guarded macro system satisfying (G1)–(G9), uniformly in the future requirement. Apply Theorem 14.3 at the exact entry Z . Its bound B_{GM} is already uniform over all stable zero exits and all exact backward requirements, so we may take $B_{\text{FR}}^\varepsilon(Z) = B_{\text{GM}}$. $\square$

15 Reset-ending first returns and effective zero successors

We now return to the deterministic stable/reset structure. The local compactness theorem Theorem 14.4 applies directly to ordinary positive first returns. The only new deterministic phenomenon is a visible step that enters a reset control and thereby returns silently to stable counter zero. We show that this case can be reduced to the same compactness theorem by a genuine auxiliary construction that stays inside the epsilon-OCA/epsilon-OCN model.

15.1 Consecutive stable zero cuts

Lemma 15.1 (Stable-zero trichotomy). *Let $u \neq \varepsilon$ be the visible factor between two consecutive stable zero cuts of the deterministic stable/reset automaton. Exactly one of the following occurs.*

1. Direct zero step: $|u| = 1$, and after the unique visible step and its forced reset closure, if any, the stable counter is zero.
2. Ordinary positive first return: after the first visible letter the stable counter is positive, all strict internal stable cuts are positive, and the final visible step reaches stable zero without entering a reset control.
3. Reset-ending positive first return: after the first visible letter the stable counter is positive, all strict internal stable cuts are positive, and the final visible letter enters a reset control whose forced silent behavior ends at stable zero.

Proof. If a reset occurs at a strict internal stable boundary, then its forced silent closure produces an earlier stable zero cut, contrary to consecutiveness. Hence all strict internal stable cuts are positive and reset-free. The final visible step either reaches stable zero directly or enters a reset control. The remaining case is the one-letter direct zero step. $\square$

15.2 Reset signatures

Fix a reset-ending factor of the form ua . Immediately before the final letter a , suppose the deterministic machine is in stable control p at height $h > 0$. Let the a -rule have deterministic counter effect j , enter reset control s , and let

$$m = \text{per}_s, \quad r \equiv h + j \pmod{m}, \quad q = \text{goto}_s(r).$$

The finite tuple

$$\sigma = (p, a, j, s, m, r, q)$$

is called the *reset signature*. There are only finitely many such signatures. While reading the reset-free prefix u , the residue $h \bmod m$ can be carried in finite control.

15.3 A genuine reset-drain auxiliary instance

Fix a stable zero entry $Z = (p_0, H)$ and a reset signature σ as above. Construct a deterministic auxiliary automaton $A_{Z,\sigma}^{\text{reset}}$ as follows.

From the fresh stable-zero entry representing Z , the prefix phase first copies the deterministic zero-regime visible step of A whenever its forced closure reaches a stable positive configuration without entering a reset. Thereafter it copies only the positive-regime, reset-free stable dynamics of A , while carrying the current counter residue modulo m in finite control. Every visible transition entering any reset control before the designated final marker is omitted, and no zero-regime transition is provided from a positive-copy control. Consequently, if the copied run reaches counter zero before the marker, it is stuck and cannot later reach the private drain exit. From the copy of the designated pre-reset control p , and only in the positive regime and in residue classes satisfying

$$h + j \equiv r \pmod{m},$$

a fresh visible marker $\#$ with counter effect 0 enters a private drain control. The marker deliberately does not reproduce the original update j : the reset discards the post- a value $h + j$, while the only information about that value needed to select the correct reset target is its residue, which is already enforced by the guard above. Keeping effect 0 preserves the pre-reset height h so that the drain exposes it as an ordinary visible first return. At the drain control the fresh letter d is available only in the positive regime, decrements the counter by one, and remains in the drain control. Thus every run reaching the drain-zero exit is automatically a strictly positive first return, and its first return to counter zero occurs after a word

$$u \# d^h,$$

where the corresponding original factor ua has reset signature σ . The zero drain control is declared stable and is regarded as the fixed auxiliary exit representing the stable reset target q .

The net side realizes the final visible a semantically. Construct $N_{a \rightarrow \#}$ by retaining all original epsilon transitions and, for every original visible transition

$$x \xrightarrow{a,w} y,$$

adding the copied transition

$$x \xrightarrow{\#,w} y.$$

All original visible transitions on the whole alphabet Σ are retained, independently of the eventual witness u , and d is a zero-effect self-loop at every N-control.

Lemma 15.2 (Copied-marker/drain identity). *For every epsilon-saturated envelope K ,*

$$T_{\#}^{\varepsilon}(K) = T_a^{\varepsilon}(K), \quad T_d^{\varepsilon}(K) = K.$$

Consequently, whenever $u\#d^h$ is an auxiliary reset-drain first return,

$$T_{u\#d^h}^\varepsilon(H) = T_{ua}^\varepsilon(H).$$

Proof. A weak $\#$ path in the auxiliary net is exactly a path $\varepsilon^* \# \varepsilon^*$ whose unique visible edge is a copied a -edge with the same source, target, and counter effect as in N ; replacing it by the original a -edge gives a bijection preserving requirements and exit credit. The fresh letter d is a zero-effect self-loop at every N-control, so on an already epsilon-saturated envelope its weak transformer is the identity. $\square$

Lemma 15.3 (Reset-auxiliary uniformity certificate). *For fixed machines A, N , a concrete stable zero entry $Z = (p_0, H)$, and a reset signature σ , the auxiliary pair*

$$(A_{Z,\sigma}^{\text{reset}}, N_{a \mapsto \#})$$

and its exact entry envelope are effectively determined before any reset-ending witness, pre-reset height, or future backward requirement is chosen. In particular, the height h is not compiled into the machines: it is the runtime counter value, while its residue modulo m is tracked in finite control. Thus the guarded-macro palette and the compactness bound for the auxiliary pair are uniform over all factors of signature σ and all inherited suffix requirements.

Proof. The deterministic auxiliary control graph is obtained from the finite stable/reset presentation of A by product with the residue set $\mathbb{Z}/m\mathbb{Z}$, deletion of reset-entering prefix edges, and addition of the fixed marker/drain gadget. The net auxiliary is obtained from N by copying the finitely many a -edges to $\#$ and adding the fixed d -loops. All of these operations depend only on A, N, Z, σ . The exact entry is Z with envelope H . Neither the literal prefix u , nor its runtime height h , nor the later backward vector R enters the construction. The suffix-parametric guarded-macro theorem therefore computes one bound for this fixed auxiliary instance, uniformly in R . $\square$

For an auxiliary reset-drain first return put

$$\text{rcost}(u\#d^h) = |u| + 1.$$

Thus d -letters have zero reset cost, and the reset cost is exactly the visible length of the corresponding original factor ua .

Lemma 15.4 (Residual-minimal reset comparison). *Let a reset-ending factor ua of signature σ occur between consecutive stable zero cuts of a globally shortest word in $L(A) \setminus L(N)$, and let R be the exact weak backward requirement of the unchanged suffix after the reset. If the pre-reset height is h , then $P = u\#d^h$ is reset-cost-minimal among all auxiliary positive first returns Q from the same exact entry to the drain-zero exit satisfying*

$$\text{Rej}(T_Q^\varepsilon(H), R).$$

Proof. By Theorem 15.2, the exit envelope of P is exactly the original envelope after ua , so it rejects the requirement R . Every auxiliary positive first return to the drain-zero exit has the form $Q = u'\#d^{h'}$. The residue guard on $\#$ guarantees that the original word $u'a$ has the same reset signature σ and therefore reaches the same stable zero control q after the forced reset. Again by Theorem 15.2, its net envelope is exactly the exit envelope of Q .

If such a Q rejected R and had $\text{rcost}(Q) < \text{rcost}(P)$, replacing ua by $u'a$ and keeping the original suffix would give a strictly shorter word accepted by A and rejected by N , contradicting global shortestness. Hence P is reset-cost-minimal. $\square$

Theorem 15.5 (Effective reset-ending first-return bound). *For every concrete stable zero entry Z and reset signature σ there is a total computable bound $B_{\text{reset}}^\varepsilon(Z, \sigma)$ such that every reset-ending factor of signature σ occurring between consecutive stable zero cuts of a globally shortest counterexample has visible length at most this bound.*

Proof. Fix such a factor and its suffix requirement R . Among all auxiliary positive first returns from the concrete auxiliary entry to the drain-zero exit that reject R , choose one, say Q_* , of minimum ordinary visible length. It exists because $P = u\#d^h$ is one such return. The suffix-parametric form of robust compactness, Theorem 14.4, applied to the fixed auxiliary pair $(A_{Z, \sigma}^{\text{reset}}, N_{a \rightarrow \#})$, gives a computable number $\hat{B}_{\text{FR}}^\varepsilon(Z, \sigma)$ such that

$$|Q_*| \leq \hat{B}_{\text{FR}}^\varepsilon(Z, \sigma).$$

By Theorem 15.3, this number is computed uniformly from A, N, Z, σ and does not depend on the chosen factor, its pre-reset height, or R .

By Theorem 15.4, P is reset-cost-minimal. Therefore

$$|ua| = \text{rcost}(P) \leq \text{rcost}(Q_*) \leq |Q_*| \leq \hat{B}_{\text{FR}}^\varepsilon(Z, \sigma).$$

Taking the last quantity as $B_{\text{reset}}^\varepsilon(Z, \sigma)$ proves the claim. Notice that this argument does not assert that the particular drain word $u\#d^h$ is ordinary-length-minimal; this is exactly why the separate reset cost is introduced. $\square$

15.4 Effective next-zero successors

For a concrete stable zero node $Z = (p, H)$, let $B_{\text{ord}}^\varepsilon(Z)$ be the suffix-independent ordinary first-return bound from Theorem 14.4. For each syntactically feasible reset signature σ of A , let $B_{\text{reset}}^\varepsilon(Z, \sigma)$ be the bound from Theorem 15.5; if there is no such signature, the corresponding maximum below is defined as 0. Define

$$B_0^\varepsilon(Z) = \max\left(1, B_{\text{ord}}^\varepsilon(Z), \max_{\sigma} B_{\text{reset}}^\varepsilon(Z, \sigma)\right),$$

up to the fixed boundary-letter convention already absorbed into the auxiliary constructions.

Theorem 15.6 (Effective next-zero successors). *From every concrete stable zero node $Z = (p, H)$ one can effectively compute a finite set*

$$\text{Succ}_0^\varepsilon(Z)$$

of exact next-stable-zero successors, complete for every shortest word in $L(A) \setminus L(N)$ passing through Z .

Proof. Enumerate every visible word u with $1 \leq |u| \leq B_0^\varepsilon(Z)$. Simulate the deterministic stable/reset machine exactly, retaining u iff it is a literal factor from the current stable zero configuration to the next stable zero configuration and contains no earlier stable zero cut. By Theorems 15.1 and 15.5, every consecutive-zero factor of a shortest counterexample is covered by the direct, ordinary, or reset-ending bound and therefore appears in this finite enumeration.

For each retained word compute the exact target envelope

$$H' = T_u^\varepsilon(H)$$

using Theorems 12.3 and 12.5, and record the target stable control, exact envelope, literal word, and visible cost. All steps are effective and the enumeration is finite. $\square$

16 The finite stable-zero tree

We next organize the finitely branching zero-successor relation by an extended Dickson order.

Definition 16.1 (Stable zero state). A stable zero state is a pair

$$Z = (p, H),$$

where p is a stable deterministic zero control and $H \in D^{Q_N}$. For equal deterministic control define

$$(p, H) \preceq_0 (p, K) \iff H \leq K$$

componentwise. States with different deterministic controls are incomparable.

Lemma 16.2 (Extended Dickson property). *For every finite set of N -controls, D^{Q_N} is a well quasi-order under componentwise order.*

Proof. Take an infinite sequence of vectors in D^{Q_N} . There are only finitely many masks describing the coordinates equal to ω , so pass to an infinite subsequence with one fixed omega mask. On the remaining coordinates, $D \setminus \{\omega\} = \{\perp\} \cup \mathbb{N}$ is order-isomorphic to $\mathbb{N}$ via $\perp \mapsto 0$ and $n \mapsto n+1$. Dickson's lemma gives two comparable vectors on those coordinates, while the omega coordinates agree. $\square$

The key deletion statement uses only monotonicity of weak N semantics, not arithmetic involving ω and ∞ .

Lemma 16.3 (Stable zero-cut domination). *Let $i < j$ be two stable zero cuts on a rejected deterministic accepting run. Suppose the deterministic stable zero control is the same at both cuts and that the exact envelopes satisfy*

$$H_i \leq H_j.$$

Then deleting the visible factor between the two cuts preserves deterministic acceptance and N -rejection.

Proof. The deterministic run rejoins at the identical physical configuration $(p, 0)$, so its entire later behavior, including every forced reset closure, is unchanged. Let v be the inherited visible suffix after cut j . If v were accepted from H_i , monotonicity and $H_i \leq H_j$ would imply acceptance from H_j , contradicting rejection of the original word. Thus the shortened word remains rejected. $\square$

Starting from the initial stable zero context, generate the raw successor tree using Theorem 15.6. Prune a candidate node (p, H') if an ancestor on the same branch has the same deterministic control p and an envelope $H \leq H'$.

Theorem 16.4 (Finite stable-zero tree). *The pruned zero-successor tree $\mathcal{T}_0^\varepsilon$ is finite and effectively constructible. Moreover, the complete stable-zero sequence of every shortest counterexample embeds literally as one root-to-node path in $\mathcal{T}_0^\varepsilon$.*

Proof. Every node has finitely many children by Theorem 15.6. An infinite unpruned branch would be a bad sequence for the wqo of Theorem 16.2, which is impossible. Hence the finitely branching tree has no infinite branch and is finite by König's lemma. Exhaustive generation therefore terminates.

For completeness, consider the literal stable-zero sequence of a shortest counterexample. Each consecutive factor appears among the raw successors by Theorem 15.6. If a later literal node were pruned by an earlier dominated node on that same path, Theorem 16.3 would give a strictly shorter counterexample, contradicting shortestness. $\square$

Let

$$D_0^\varepsilon$$

be the maximum root-to-node edge depth of $\mathcal{T}_0^\varepsilon$, and let

$$L_0^\varepsilon$$

be the maximum sum of visible edge costs along a root-to-node path. Both are total computable integers. Every shortest counterexample has at most $D_0^\varepsilon + 1$ stable zero cuts, and its visible prefix through the last stable zero cut has length at most L_0^ε .

17 Terminal tails and the global hard-inclusion bound

It remains to control the suffix after the last stable zero cut. Such a suffix cannot contain a reset: every reset ends at stable counter zero and would create a later stable zero cut.

17.1 A syntactic terminal auxiliary automaton

Fix a stable zero node $Z = (p, H)$ from $\mathcal{T}_0^\varepsilon$. Construct a deterministic auxiliary automaton A_Z^{term} syntactically. From a fresh zero start representing p , copy exactly those zero-regime visible transitions whose target is a stable positive configuration. Omit transitions entering reset controls or returning immediately to stable zero. For every stable control q , create a positive copy q^+ and copy every positive-regime visible transition whose target is stable; transitions entering reset controls are omitted. If a copied decrement reaches zero, the run is stuck because no zero-regime original transitions are available inside the positive copy.

From every copied accepting stable control f^+ add a fresh marker $\#$ *only in the positive regime*, with counter effect 0, into a fresh stable drain control. At the drain control, the fresh letter d is available only in the positive regime and decrements the counter by one; the fresh letter $\$$ is available only in the zero regime and enters the unique accepting control. No zero-regime $\#$ rule is present at any positive-copy state. Hence a copied decrement that reaches zero before the marker is truly stuck, while an accepted run reaches zero for the first time during the private drain. Thus the accepted words are exactly

$$u \# d^h \$,$$

where u is a reset-free positive terminal tail ending at an original accepting stable control at height $h > 0$.

On the N side retain all original transitions and add zero-effect self-loops on $\#, d, \$$ at every control. Both auxiliary machines are constructed uniformly from (A, N, Z) alone; neither the particular terminal tail, its ending height, nor any future backward requirement is used in their control graphs.

Lemma 17.1 (Fresh identity letters). *If K is epsilon-saturated and $x \in \{\#, d, \$\}$ is one of the fresh identity letters, then*

$$T_x^\varepsilon(K) = K.$$

Consequently,

$$T_{u\#d^h\$}^\varepsilon(H) = T_u^\varepsilon(H).$$

Proof. A weak x path has the form $\varepsilon^* x \varepsilon^*$ and the visible x -edge is a zero-effect self-loop at every control. Hence its transformer is $C_\varepsilon \circ \text{Id} \circ C_\varepsilon$, which equals the identity on an already epsilon-saturated envelope. Iterating over the fresh suffix gives the second equality. $\square$

Theorem 17.2 (Effective terminal-tail bound). *For every concrete zero node Z , one can compute an integer $B_{\text{tail}}^\varepsilon(Z)$ such that, if there exists any accepting terminal tail from Z rejected by N , then there exists one of visible length at most $B_{\text{tail}}^\varepsilon(Z)$.*

Proof. Choose a shortest counterexample of the auxiliary terminal instance. It has shape $u\#d^h\$$. The prefix $u\#d^h$ starts at stable zero, is positive after its first visible letter, contains no internal reset or stable zero cut, and reaches zero for the first time on the final d . It is therefore an ordinary positive first-return corridor. By Theorem 14.4, its visible length is bounded by a total computable integer depending only on the concrete auxiliary entry. By Theorem 17.1, rejection of the auxiliary word is equivalent to rejection of the original tail u from the same exact envelope H . Taking the first-return bound as $B_{\text{tail}}^\varepsilon(Z)$ proves the claim. $\square$

Define

$$B_{\text{tail},\max}^\varepsilon = \max_{Z \in \mathcal{T}_0^\varepsilon} B_{\text{tail}}^\varepsilon(Z)$$

and

$$B_{\text{global}}^\varepsilon = L_0^\varepsilon + B_{\text{tail},\max}^\varepsilon.$$

Both are computable because the zero tree is finite.

Theorem 17.3 (Global small counterexample). *If*

$$L(A) \not\subseteq L(N),$$

then there exists

$$w \in L(A) \setminus L(N)$$

with

$$|w| \leq B_{\text{global}}^\varepsilon.$$

Proof. Take a shortest counterexample w . Let x be the prefix ending at its last stable zero cut, with zero node $Z = (p, H)$, and write $w = xu$. By Theorem 16.4, $|x| \leq L_0^\varepsilon$.

If $u = \varepsilon$, the claim follows immediately. Otherwise u is a reset-free positive terminal tail from Z . By Theorem 17.2, there exists an accepting tail v from the same deterministic physical configuration and the same exact N envelope H , rejected by N , with $|v| \leq B_{\text{tail}}^\varepsilon(Z)$. Thus xv is again in $L(A) \setminus L(N)$ and

$$|xv| \leq L_0^\varepsilon + B_{\text{tail},\max}^\varepsilon = B_{\text{global}}^\varepsilon.$$

$\square$

Proof of Theorem 1.4. Compute $B_{\text{global}}^\varepsilon$ and enumerate all visible words of lengths

$$0, 1, \dots, B_{\text{global}}^\varepsilon.$$

The length-0 case is included explicitly. For each word, simulate the unique deterministic stable/reset run of A and compute the exact weak N envelope from Theorem 12.3; final N-acceptance is tested after trailing epsilon saturation. By Theorem 17.3, the inclusion fails if and only if one of these finitely many words belongs to $L(A) \setminus L(N)$. $\square$

18 Reverse inclusion

We now decide $L(N) \subseteq L(A)$. This direction does not use the small- witness theorem.

18.1 A marked noninclusion reduction

Introduce a fresh visible endmarker $\dashv \notin \Sigma$. First totalize the visible stable semantics of A by sending every missing stable visible rule to one fresh stable dead control, with dead self-loops on all original visible letters in both sign regimes.

Construct a marked deterministic rejection automaton A_{bad} by adding $\dashv$ -rules from stable controls: an original accepting stable control reads $\dashv$ into a nonaccepting sink GOOD, whereas every nonaccepting stable control, including the dead control, reads $\dashv$ into an accepting sink BAD. The marked sinks have no outgoing transitions.

Construct N_{end} by retaining all original transitions and, from every original accepting N-control q , adding

$$q \xrightarrow{\dashv, 0} f_{\text{end}},$$

where f_{end} is the unique marked accepting control.

Lemma 18.1 (Marked noninclusion).

$$L(N) \not\subseteq L(A)$$

if and only if there exists $w \in \Sigma^$ such that both marked systems have accepting runs on $w \dashv$.*

Proof. The totalized deterministic stable/reset machine has one unique stable endpoint after every visible word. Hence A_{bad} accepts $w \dashv$ exactly when the original A rejects w . On the net side, arbitrary trailing epsilon moves may occur before the endmarker, so N_{end} accepts $w \dashv$ exactly when N accepts w . The case $w = \varepsilon$ is represented by the one-letter marked word $\dashv$. $\square$

18.2 Reset decompression

For a reset control s of period p , replace the semantic reset

$$(s, n) \xrightarrow{\varepsilon^*} (\text{goto}_s(n \bmod p), 0)$$

by finite micro-controls

$$s[0], \dots, s[p-1].$$

At positive counter add

$$s[i] \xrightarrow{\varepsilon, -1} s[(i+1) \bmod p],$$

and at zero add

$$s[i] \xrightarrow{\varepsilon, 0} \text{goto}_s(i).$$

Starting from height n , exactly n silent decrements occur and the zero exit is taken in phase $n \bmod p$.

Lemma 18.2 (Commutation with N-epsilon moves). *Inside the asynchronous product, N-epsilon moves commute with the decompressed A-reset microsteps without changing component endpoints or visible projection.*

Proof. An N-epsilon move changes only the N-control and N-counter, while an A-reset microstep changes only the A-control and A-counter. Neither consumes visible input. Their feasibility conditions therefore concern disjoint counters, so adjacent independent moves may be swapped. Repeating the swap moves all N-epsilon steps to either side of the finite A-reset microsequence while preserving both component endpoints. $\square$

18.3 The asynchronous marked product

A genuine product control stores one control of N_{end} and one stable or reset-micro control of A_{bad} . The two main counters are denoted by x_N and x_A .

N-epsilon moves change only (q_N, x_N) . Decompressed A-epsilon moves change only (q_A, x_A) . An original visible letter synchronizes one chosen N-edge with the unique deterministic A-edge in the current A sign regime. The endmarker $\dashv$ is synchronized in the same way. By Theorem 18.2, allowing arbitrary interleavings of the two silent components introduces no new visible behavior.

Lemma 18.3 (Product/run correspondence). *A marked accepting product control is reachable if and only if there exists $w \in L(N) \setminus L(A)$.*

Proof. Projecting a product run gives weak component runs with the same visible word $w \dashv$. Conversely, interleave the silent moves of any two such component runs and synchronize their visible letters; Theorem 18.2 handles reset microsteps. Apply Theorem 18.1. $\square$

18.4 Compilation to one zero-test VASS

The low-level construction is given in Appendix F; we record here only the correctness interfaces used by the theorem. A positive-regime synchronized edge with effects (d_N, d_A) is compiled through a private intermediate control as

$$(d_N, -1) \quad \text{then} \quad (0, d_A + 1),$$

so ordinary nonnegativity enforces $x_A > 0$ and the private control makes the gadget atomic. Positive reset microsteps already have effect $(0, -1)$.

All zero-regime semantic edge types are serialized through one common gate. Each type e has an ordinary tag counter t_e : its source transition records e , one distinguished transition g_0 tests only $x_A = 0$, and the tagged post-gate branch consumes t_e , performs the fixed ordinary vector update of e , and returns to a genuine product control.

Lemma 18.4 (Single-gate exactness). *A complete gate path for a semantic zero edge e reaches its genuine target if and only if the edge is taken at its proper source with $x_A = 0$ and its ordinary counter update is feasible. At every genuine product control all tag counters are zero. Thus g_0 is the unique zero-test transition.*

Proof. The unique tag identifies the chosen edge; all other completion branches are disabled by ordinary nonnegativity. If the vector update is infeasible the run is trapped inside the private gadget. A successful completion consumes the tag and restores the all-zero tag invariant. See Appendix F for the explicit transition schema. $\square$

From the marked accepting product control, a private cleanup phase supplies ordinary decrement loops for every counter and one final zero-effect edge to a fixed target.

Lemma 18.5 (Cleanup equivalence). *Some marked accepting product configuration with arbitrary nonnegative counters is reachable if and only if the fixed all-zero cleanup target is exactly reachable.*

Proof. Drain all counters before taking the final edge. Conversely cleanup is reachable only from marked acceptance, and taking the final edge early cannot reach the queried all-zero configuration. $\square$

Exact reachability for VASS/Petri nets with one zero test (equivalently, one inhibitor arc in the standard formulation) is decidable [10, 11]. Our compiled system matches that model exactly: g_0 is the single zero-test transition, all remaining counters are ordinary VASS counters, and every other transition is an ordinary vector addition.

Proof of Theorem 1.5. By Theorems 18.3 to 18.5, $L(N) \not\subseteq L(A)$ if and only if the fixed all-zero cleanup target is reachable in the compiled VASS with one zero test. The latter reachability problem is decidable, so $L(N) \subseteq L(A)$ is decidable. $\square$

Proof of Theorem 1.3. By Theorem 1.4, the inclusion $L(A) \subseteq L(N)$ is decidable. By Theorem 1.5, the reverse inclusion $L(N) \subseteq L(A)$ is decidable. Therefore one decides $L(A) = L(N)$ by deciding both inclusions and accepting exactly when both hold. $\square$

19 Conclusion

Language equality between deterministic zero-testing one-counter automata and nondeterministic one-counter nets is decidable in the standard real-time setting and remains decidable under the epsilon-transition conventions of Theorem 1.3. The common mechanism is a shortest-counterexample compactness theory organized around exact residual semantics and deterministic zero returns. Silent net behavior is absorbed by a finite threshold-affine weak-macro calculus, while deterministic silent behavior is handled by stable/reset normalization.

The resulting bounds are effective but intentionally complexity-agnostic. They combine well-founded rank recursions, finite Ramsey and Dickson arguments, Presburger checks, and one-zero-test reachability subroutines. Determining useful complexity bounds is left open.

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A Real-time effectivity ledger

The real-time global small-witness theorem needs a *computable* witness bound. The following list records the external or finite-search ingredients used to compute that real-time bound. The epsilon extension does *not* use the finite-path statement in item 1; its corresponding effectivity is supplied instead by the exact weak-envelope construction and guarded-macro ledgers in Appendices D and E.

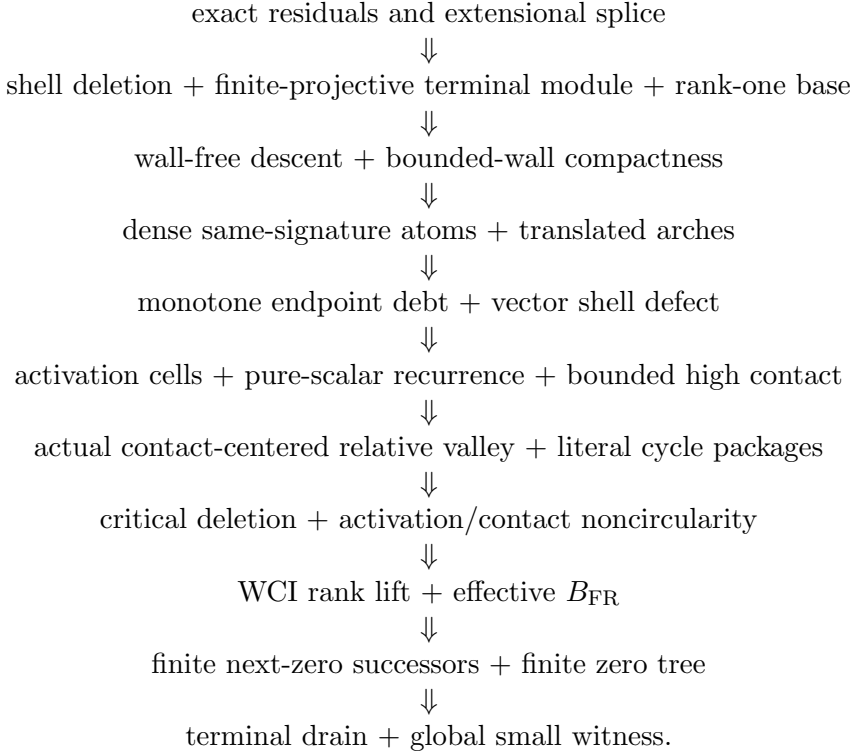
1. Exact net transformers T_x and S_x for a fixed word are finite dynamic programs over the finitely many paths of length $|x|$.
2. Support, owner frames, primitive synchronized simple cycles, exact wall signatures, and bounded-context carrier colors are finite and effectively enumerable.
3. Rank-one recurrent owner switching is compiled to a VASS with one zero-tested scalar (Proposition 5.17); exact reachability for that class is decidable [10, 11].
4. Finite-projective systems reduce to exact 2-VASS reachability and shortest representative search (Proposition 5.13).

5. The dense-wall constants are explicit: $K_{\text{dense}} = 4|\mathcal{W}| + 1$, and a requested number of dense atoms is forced by a computable linear wall-count threshold.
6. Every finite Ramsey use is on an explicitly finite coloring (for example side $\times$ envelope coordinate in Lemma 5.23); only finite Ramsey numbers are required.
7. Every tracked integer observable has a computable one-macro drift bound; first exit from a cutoff occurs in a finite band.
8. The effectivity proof avoids an uncontrolled infinite Dickson subsequence by the reserve split (31); below the explicit projective cutoff the normalized profile space is finite.
9. The translated-arch activation bridge is quantitative: bounded-cell refinement fixes the finite tuple $(c, \delta, d, p, \eta, u, v)$; the pure-scalar survivor fixes (d, λ, η, K) ; the bounded-contact state, requirement, and counter lie in finite computable sets. Hence the contact/relative-valley transition introduces no infinitary choice.
10. The proof-state recursion uses the explicit finite stage set in Table 1 and the lexicographic measure (ν, ρ, s) .
11. Next-zero successors are obtained by literal word enumeration up to the computed first-return bound; exact envelopes are computed by dynamic programming.
12. The pruned zero-state tree is effectively generated. It terminates because it is finitely branching and every branch is a Dickson bad sequence.
13. Terminal tails reduce to another concrete first-return instance by the fresh-marker drain construction, after which a finite maximum is taken over the finite zero tree.

B Real-time canonical dependency graph and excluded proof routes

The real-time compactness proof uses only the corrected canonical dependency chain below. The epsilon theorem invokes this core only through the guarded-macro transfer theorem of Section 14.

Exploratory routes that were later superseded are not premises of any theorem in this paper.



For referee clarity, several historically tempting routes are *not* dependencies of the theorem.

Excluded shortcut	Why it is insufficient	Canonical replacement
Uniform separator “gap spending”	the rejection margin is monotone but need not decrease as a consumable budget	endpoint domination, anchored shell deletion, and rank descent
Global compilation of recurrent wall equality into one lower-rank machine	unboundedly many exact wall equalities may require independent equality checks	fixed- K compilation plus dense same-signature returns and translated arches
Treating 2-VASS flattening as trace preserving	reachability endpoints may be represented by a different input word	actual-trace cycle erasure for literal deletions; VASS alternatives only for endpoint replacements
Strong arbitrary lower-rank exact-slice/Pareto oracle	stronger than the shortest-witness induction can justify at every rank	bounded-context domination
Infinitary Dickson extraction for normalized endpoint profiles	abstract WQO alone does not yield the desired quantitative threshold without growth control	WCI(ρ) plus responsible-escape alternative
Nested-zero-test shortcut at wall boundaries	simultaneous-zero compatibility of high-rank wall tests with lower-rank tests is not established	explicit scalar/projective reserve split and finite profile boxes
		extensional fixed- K wall handling and literal high-shell deletion

C Real-time trace-preservation checklist

Every use of cycle or VASS machinery in the real-time compactness proof falls into one of the following two audited categories. The epsilon transfer preserves the same distinction by retaining concrete weak skeletons, as recorded in Appendix E.

1. **Literal operation on the observed witness.** The factor to be removed is an actual contiguous cycle occurrence (or a finite sparse set of such occurrences) found by cycle erasure on the observed synchronized trace. Deterministic safety is verified on that literal word. No alternative path scheme is substituted silently.
2. **Constructive endpoint replacement.** A reachability theorem may return a different word. The proof records its exact deterministic endpoint and exact N exit envelope and invokes Lemma 5.5. Proposition 5.13 and the bounded-projective terminal branches are of this form.

The support-two southwest reduction and the A -neutral package in Section 5.14 belong to category 1. Finite-projective compression belongs to category 2. No theorem crosses from category 2 to a literal deletion claim without an independent trace-faithful extraction.

D Effective weak-envelope construction

This appendix supplies the algorithmic details behind Theorems 12.3 and 12.5. The purpose is not to eliminate silent moves syntactically, but to compile their exact effect on upper envelopes and backward requirements into a finite collection of clauses.

D.1 Effective finite and pump enumeration

Fix p, q and a visible letter a . Enumerate every simple epsilon-prefix, one a -edge, and every simple epsilon-suffix. Each resulting skeleton π contributes the exact finite clause

$$c \mapsto c + \text{eff}(\pi) \quad (c \geq \text{req}(\pi)).$$

For pumping, enumerate a simple positive epsilon cycle together with simple connectors from the source, through the visible edge, and to the target, with the cycle on either silent side. For a lasso $\rho C \tau$, the pump threshold is $\text{req}(\rho C)$. The proofs of Theorems 12.1 to 12.3 show that these finite families are complete: every finite maximizing run is dominated by a simple skeleton, while every pumpable run contains one of the enumerated positive-cycle witnesses. Hence both the finite clause palette and the exact onset of the pumping branch are effectively computable.

D.2 Exact propagation from an omega source

If $H(p) = \omega$, the set of reachable finite counters at p is unbounded. For any finite clause with threshold r , some reachable finite source counter is at least r , and applying the fixed clause to arbitrarily large reachable source counters produces arbitrarily large target counters. Thus the target value is ω . No congruence or eventual-density assumption is used.

D.3 Backward transformer

Let R be a target requirement vector. For a finite simple skeleton $\pi : p \Rightarrow_a q$ with $R(q) < \infty$, the least source credit that both executes π and leaves enough target credit is

$$b(\pi, q, R) = \max\{\text{req}(\pi), R(q) - \text{eff}(\pi), 0\}.$$

A pump witness leading to a target with finite $R(q)$ contributes its pump entry threshold itself: after the positive cycle is executable, enough repetitions can manufacture any prescribed finite target credit. Targets with $R(q) = \infty$ contribute no candidate. Therefore

$$S_a^\varepsilon(R)(p)$$

is the minimum over a finite effectively computable candidate set, or ∞ when that set is empty. Upward closure of OCN feasibility then gives

$$\text{Acc}(T_a^\varepsilon(H), R) \iff \text{Acc}(H, S_a^\varepsilon(R)),$$

including the ω case. This also proves that the initial saturation H_ε and terminal requirement R_ε are effectively computable by the same finite epsilon-path/cycle analysis with the visible edge omitted.

E Macro-transfer and constant-rescaling ledger

We collect here the quantitative interface between the guarded-macro transfer theorem Theorem 14.3 and the weak-macro system of Section 13. The real-time argument is contained earlier in this paper: owner/provenance and finite-projective setup in Sections 5.2–5.4, wall and translated-arch localization in Sections 5.5–5.9, threshold/contact and literal-trace analysis in Sections 5.10–5.15, the structural rank lift in Theorem 5.46, and the quantitative recursion in Theorem 5.47. The point of this appendix is to isolate every place where those arguments used unit real-time semantics and to give its weak-macro replacement.

Let $\mathcal{M}$ be the finite palette of finite weak macro types after refining by source, target, visible letter, threshold, effect, and concrete simple skeleton identity. Let $\mathcal{P}$ be the finite palette of pump witness types. Put

$$R_* = \max(\{r_e : e \in \mathcal{M}\} \cup \{r_p : p \in \mathcal{P}\} \cup \{0\})$$

and

$$E_* = \max(\{|\delta_e| : e \in \mathcal{M}\} \cup \{0\}).$$

Both are effectively computable.

E.1 Per-step observable bounds

On a fixed finite regime:

- one N/N projective difference changes by at most $2E_*$;
- one anchored A/N relative coordinate changes by at most $E_* + 1$, since the deterministic stable step has unit effect;
- a common anchored N-credit observable changes by at most E_* ;
- support, provenance, owner, and omega masks range over finite sets.

Thus one conservative universal first-exit constant is

$$D_* = \max\{1, 2E_* + 2\}.$$

Whenever a tracked integer observable first exits $[-M, M]$, it lies in the finite band

$$M < |z| \leq M + D_*.$$

This is the only change required in first-exit localization.

E.2 Threshold events

Every finite macro enabling change occurs at one of the finitely many thresholds $r_e \leq R_*$. Every pump enabling change occurs at one of the finitely many pump thresholds $r_p \leq R_*$. Hence a first semantic-regime split is localized either by a finite support/omega event or by a source counter in a bounded band around one of these thresholds. For a recurrent threshold contact the color records this bounded source credit and finite mode data, but not the numerical separator margin; by Theorem 13.5 the resulting contact alphabet is finite and independent of the future backward requirement.

E.3 Projective cutoffs

Any real-time estimate that bounded the accumulated effect along a simple synchronized path by the number of vertices is replaced by the same path length multiplied by the appropriate macro differential bound. In particular, if a simple residual path contains at most $|V| - 1$ macro edges, its N/N differential contribution has absolute value at most

$$(|V| - 1) 2E_*.$$

Thus every explicit projective-spread cutoff remains computable after one finite rescaling.

E.4 Transfer certificate

The structural use of these constants is summarized in the proof of Theorem 14.3. No additional semantic assumption is hidden in this appendix: all owner/provenance, wall/contact, WQO, terminal-oracle, and powered-route requirements are exactly (G1)–(G9). The purpose of the present ledger is only to make the finite rescalings R_* , E_* , and D_* explicit.

E.5 Literal-trace interface

The transferred proof may delete only cycles/packages extracted from the actual observed trace. Accordingly, each macro edge used by an owner/provenance strand stores one concrete simple weak skeleton. Consecutive selected macro edges concatenate to an actual N micro-run. A nonempty synchronized macro cycle therefore consumes a nonempty contiguous visible factor. Alternative VASS paths may be used for decidability or fixed-route checks, but never as a substitute for the literal observed trace in a shortest-word deletion.

F Formal one-zero-test product gadget

This appendix records the low-level compilation used in Section 18. It is included to make the exact match with the external one-zero-test reachability theorem transparent.

F.1 Semantic product edges

Before compilation, a semantic product edge records:

- one source product control and one target product control;
- the chosen N transition, when the edge is visible;
- the unique deterministic A rule in the current sign regime;
- the ordinary vector effect on (x_N, x_A) ;
- whether the A source guard is positive or zero.

N-epsilon and positive reset-decrement edges are already ordinary VASS transitions and need no special treatment.

F.2 Positive-source gadget

For a positive-source semantic edge with effect (d_N, d_A) and $d_A \in \{-1, 0, +1\}$, introduce one private intermediate control and replace the edge by

$$(d_N, -1) \quad \text{followed by} \quad (0, d_A + 1).$$

The first transition is enabled exactly when the original N update is feasible and $x_A \geq 1$. The second transition only adds a nonnegative amount to x_A . No other transition is incident from the private intermediate control, so the gadget is atomic with respect to all asynchronous behavior.

F.3 Unique zero-test gate

Let E_0 be the finite set of zero-source semantic product-edge types. For each $e \in E_0$ introduce an ordinary tag counter t_e , initially zero. Every genuine product configuration has all tags zero.

For an edge e from genuine source s_e to genuine target t_e^{ctl} :

1. an ordinary source transition from s_e enters the common pre-gate control and increments only the tag counter t_e ;
2. the common pre-gate has one outgoing distinguished transition g_0 , enabled exactly when $x_A = 0$, with zero vector effect;
3. the common post-gate has one completion branch per semantic edge type; branch e decrements the tag t_e , performs the original ordinary vector update of e , and enters t_e^{ctl} .

All unchosen tags are zero, so their completion branches are disabled by ordinary VASS nonnegativity. The pre/post-gate controls expose no product, epsilon, visible, endmarker, or cleanup transitions other than the gadget moves just listed.

If the ordinary N component of the semantic update is infeasible, a run may cross g_0 but cannot complete the chosen branch and is trapped away from all genuine accepting controls. Hence incomplete gate attempts create no spurious accepting reachability.

It follows that a genuine source-to-target zero-edge simulation exists if and only if the chosen semantic edge is at its proper source, $x_A = 0$, and its ordinary vector update is feasible. The compiled system therefore has exactly one zero-test transition, namely g_0 .

F.4 Reset decompression and asynchronous commutation

For a reset state of period p , the phase controls

$$s[0], \dots, s[p-1]$$

perform positive silent decrements

$$s[i] \xrightarrow{\varepsilon, -1} s[i+1 \bmod p]$$

until $x_A = 0$, followed by the appropriate zero exit. N-epsilon steps modify only x_N and the N control; reset microsteps modify only x_A and the A control. Since neither consumes visible input, any interleaving of these two families commutes to an interleaving with the same component endpoints and counter values. Thus decompression does not change the synchronized visible language of the product.

F.5 Exact cleanup target

After the unique marked accepting genuine product control, enter a private cleanup phase. Each ordinary counter, including x_N , x_A , and all tags, has a decrement loop. A zero-effect edge leads to one fixed target control. The reachability query is for that control with the exact all-zero vector. Taking the final edge before complete drainage does not satisfy the query, and no edge returns from cleanup to the product. Hence arbitrary nonnegative accepting counter values are equivalent to exact reachability of one fixed all-zero configuration.

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